A constructive ATLAS of finite simple groups in Lean

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Abstract

We present a constructive atlas of finite simple groups with proofs of their orders, simplicity, and structural properties. The sixteen completed families comprise cyclic groups of prime order, alternating groups, A_r, ${}^2A_r$, B_r, C_r, D_r, ${}^2D_r$, E_6, E_7, E_8, F_4, G_2, ${}^3D_4$, ${}^2B_2$, and ${}^2G_2$, over their explicitly stated simple parameter ranges. The sixteen sporadic constructions are $M_{11}, M_{12}, M_{22}, M_{23}, M_{24}$, $\mathrm{Co}_1, \mathrm{Co}_2, \mathrm{Co}_3$, $\mathrm{McL}, \mathrm{HS}, \mathrm{Suz}, J_2$, $\mathrm{Fi}_{22}, \mathrm{Fi}_{23}, \mathrm{Fi}'_{24}$, and Ru. The Tits group ${}^2F_4(2)'$ is a separate exceptional entry. The models arise from codes, lattices, forms, algebras, finite geometries, and integral Chevalley root operators. They retain natural actions, stabilizers, central quotients, and comparison maps. Uniform order arithmetic separates the constructed groups outside the exceptional isomorphisms and two equal-order patterns. Involution classes and Sylow-subgroup centers separate those remaining pairs. A combined theorem determines isomorphism by normalized labels within its specified registry. Further equivalences identify several Chevalley and earlier geometric models. The construction and comparison proofs are formalized in Lean without assuming the exhaustiveness theorem of the classification. We state the completed results, the boundaries of the comparison theorems, and the scope of the latest independent verification.

Public release: `https://github.com/Moonshine-in-Kansas/atlas`.

Development source: `https://github.com/Moonshine-in-Kansas/atlas_draft` (private). This note describes the development snapshot of 25 October; its full contents are not asserted to be in the public release.

1 Introduction

The construction side of finite simple group theory asks for concrete models together with proofs of their orders, simplicity, and characteristic structure. Codes, lattices, algebras, forms, and finite geometries provide such models and retain natural actions, stabilizers, central quotients, and maps between related objects. The Atlas project organizes these constructions into a structural library, with proofs formalized in Lean [30] using Mathlib [26].

The catalogue reported here records the development snapshot of 25 October 2026: sixteen family packages, sixteen sporadic constructions, and the exceptional derived Tits group, giving thirty-three completed entries. Formalization results 3.1, 3.2, 4.1, and 4.2 state their orders and parameter ranges. These are construction packages, not thirty-three disjoint isomorphism classes.

The classical models come from linear, Hermitian, symplectic, and quadratic spaces. The split octonion and Albert algebras give G_2 and F_4; explicit twisted root matrices give Suzuki, triality, and small Ree groups. The uniform integral Chevalley construction supplies E_6, E_7, E_8, and further realizations of the untwisted families already present. Codes, lattices, and the Conway–Parker algebra give the sporadic models, with the Conway–Wales lattice also supplying Rudvalis and the derived Tits group. Both the defining objects and their structural maps are retained.

The comparison layer has three themes: isomorphisms between construction models, exceptional isomorphisms such as $\mathrm{PSU}_4(2) \cong \mathrm{PSp}_4(3)$, and nonisomorphism of the remaining distinct labels by orders or further invariants. The uniform classification of order coincidences applies to the proved cardinalities of the constructed groups. A combined theorem packages these arguments for the typed registry specified in Section 7.8.

Classification has two complementary components: constructing the groups on the list and proving that every finite simple group occurs there. Atlas develops the first component without assuming the exhaustiveness theorem of the classification of finite simple groups (CFSG). Construction, recognition, and exhaustiveness remain distinct obligations. The actions, stabilizers, and comparison maps retained here are intended as inputs to subsequent structure and recognition theorems; Section 2 states the conditional recognition questions precisely.

The mathematical sources combine established constructions with the author's companion work. The Hall–Janko preprint [14] and the Conway–Parker algebra manuscript [13] supply source mathematics for the corresponding entries. The present article describes the models, their order and simplicity arguments, and the structural relationships preserved in their formal realization. The direct Monster-order work with Seysen [19] supplies a starting point for future formalization, not a completed entry in this release. Companion work on integral Albert algebras and the rank-26 lattice and Cayley-plane design develops the arithmetic and geometric direction [15, 17].

A broader objective is to explain why sporadic symmetries arise. Vertex operator algebras (VOAs) provide a principal direction, illustrated by the public Monster work in Section 13 and by joint work with Lam on a VOA construction of the Fischer groups [18], currently in preparation. Retaining the mathematical objects is essential for this purpose: the Fischer construction, for example, keeps its algebra, rays, and incidence geometry together with the arguments determining the group and its order. Formal verification makes these arguments and their structural maps available for checked reuse.

The name recalls the *Atlas of Finite Groups* [5]. Our initial scope is constructions and structural theorems; complete character tables and maximal-subgroup lists are longer-term projects. Section 4 describes the added geometric families, Rudvalis, and Tits. Sections 5–12 develop the linear, quadratic, Chevalley, octonion, small Ree, and earlier sporadic constructions. Section 7 also records the classical conventions and comparison results. Sections 13 and 14 distinguish further mathematical goals from the completed catalogue. Related formalization projects are discussed in Section 15. Technical implementation, declaration interfaces, logical foundations, and verification records are collected in Appendices A–D, including the complete per-group source-line dependency table.

2 The mathematical content of an entry

For a family label X, let $\mathcal{A}_X$ be an explicitly specified parameter domain. An entry consists of a defined group $G_X(\lambda)$, for $\lambda \in \mathcal{A}_X$, and proofs of

$$|G_X(\lambda)| = N_X(\lambda), \qquad G_X(\lambda) \text{ is simple}, \qquad P_X\big(G_X(\lambda)\big), \tag{2.1}$$

together with finiteness and, for the nonabelian entries, explicit noncommutativity. Here P_X describes independently defined mathematical structure: an action, a code, a lattice, an algebra, or specified maps between such objects. Neither $\mathcal{A}_X$ nor P_X may incorporate the desired order or simplicity assertion as an unproved premise. For cyclic groups of prime order, commutativity is part of the intended conclusion.

For some entries, we also compute $k_2(G)$, the number of conjugacy classes of nonidentity involutions, to distinguish nonisomorphic groups of the same order; see Section 7.7.

Here *construction* refers to a specified mathematical model and proofs of its properties. The development uses classical mathematics in Lean, including classical choice; it need not provide an executable enumeration of every group element.

2.1 The chosen object and its full symmetry group

For a symmetry-group model, equality between a generated subgroup and the full automorphism group is a mathematical theorem. In a code or lattice construction, generation or a stabilizer argument establishes this equality before the order and simplicity assertions are attached to the full group.

Central quotients come with their maps and kernels. Thus Co_0 is distinguished from Co_1, and a Hermitian isometry group from its scalar quotient. A different distinction is that between Fi_{24} and its derived subgroup Fi'_{24}. Abstract group isomorphisms, subgroup equalities, and equivariant identifications of actions are recorded with the corresponding data.

2.2 Conditional recognition and comparison

Conditional on CFSG, order determines a finite simple group up to isomorphism except for the pairs $\{\mathrm{Alt}_8 \cong \mathrm{PSL}_4(2), \mathrm{PSL}_3(4)\}$ and $\{B_r(q), C_r(q)\}$ with $r \geq 3$ and q odd [22]. These pairs are distinguished by the number $k_2(G)$ of conjugacy classes of nonidentity involutions: the first has values 2 and 1, and the second r and $\lfloor r/2 \rfloor + 1$ [36, 4]. Consequently, assuming CFSG, the pair $(|G|, k_2(G))$ determines the isomorphism type of a finite simple group. The formal B/C counts and both nonisomorphism results are described in Section 7.7; the general recognition theorem and the Alt_8 /$\mathrm{PSL}_3(4)$ involution counts are proposed work. This conditional recognition question is distinct from the unconditional comparison theorem for the specified models in Section 7.8.

For construction proofs, the relevant independence question concerns the hypotheses and dependency graph of each theorem. Background results and earlier constructions can be reused. A classification argument can also be non-circular when the property being deduced was not used in that argument, for example when it concerns smaller groups. The Monster order illustrates why precise attribution matters: Griess–Meierfrankenfeld–Segev, Corollary 3.7.3, uses substantial information about other sporadic groups [11], whereas Höhn–Seysen [19] gives a direct computational determination of the Monster and Baby Monster orders in the Leech-lattice setting. The former dependency is more specific than an invocation of the entire CFSG.

Recognition through an action or local geometry offers a natural interface: a classification argument produces the structure, and a comparison theorem identifies it with a constructed model. The actions, stabilizers, and maps retained here are intended to support such theorems.

3 Current constructions

The development catalogue records sixteen completed family packages, sixteen sporadic constructions and the exceptional derived Tits group [12]. In this article, *completed* means that the stated order, simplicity, and structural results have the recorded proofs for the specified parameter domains; it does not mean that every property of the groups has been formalized. The verification scope is described in Appendix C. Family labels count construction packages and can overlap up to isomorphism. Section 7 records the status of the comparison maps.

Formalization result 3.1 (Family packages)**.** *The following eight packages formed version 1.0; the eight additional families in the current development are stated in Section 4. Here n is the degree of an alternating group or the matrix dimension of PSL_n; r denotes Dynkin rank in the B, C, D families. For each finite field F below, put $q = |F|$.*

(i) *For every prime p, the cyclic group C_p has order p and is simple and commutative.*

(ii) *For every $n \geq 5$, the alternating group Alt_n is nonabelian simple and has order $n!/2$.*

(iii) *Let F be any finite field, put $q = |F|$, and let $n \geq 2$. The central quotient $\mathrm{PSL}_n(F) = \mathrm{SL}_n(F)/Z(\mathrm{SL}_n(F))$ has order*

$$|\mathrm{PSL}_n(F)| = \frac{q^{n(n-1)/2} \prod_{i=2}^{n} (q^i - 1)}{\gcd(n, q-1)}. \tag{3.1}$$

It is simple if and only if $(n, q) \notin \{(2, 2), (2, 3)\}$. For those admissible parameters it is also nonabelian.

(iv) *For every finite field F and rank $r \geq 1$, the projective quadratic model $B_r(F)$ has the order in* (7.1)*. It is simple exactly outside $(r, q) = (1, 2), (1, 3), (2, 2)$, and nonabelian at the simple parameters. The full nonsimple model at $(r, q) = (2, 2)$ and its derived subgroup are distinguished.*

(v) *For every finite field F and rank $r \geq 1$, the symplectic quotient $C_r(F) = \mathrm{PSp}_{2r}(F)$ has the order in* (7.1)*. It is simple exactly outside $(r, q) = (1, 2), (1, 3), (2, 2)$, and nonabelian at the simple parameters. The full nonsimple model at $(r, q) = (2, 2)$ and its derived subgroup are distinguished. The symplectic rank-zero model is trivial.*

(vi) *For every finite field F and $r \geq 4$, the projective split-orthogonal model $D_r(F)$ has the order in* (7.2) *with $\varepsilon = 1$ and is nonabelian simple in every characteristic, including $D_4(2)$.*

(vii) *For every finite field F, with $q = |F|$, let $G_2(F) = \mathrm{Aut}_F(\mathbb{O}_{\mathrm{split}}(F))$ be the full group of multiplication-preserving linear automorphisms of the constructed split octonion algebra. Then*

$$|G_2(F)| = q^6(q^6 - 1)(q^2 - 1). \tag{3.2}$$

It is simple if and only if $q > 2$, and noncommutativity is proved for every q. At $q = 2$, an explicitly defined normal subgroup has order 6048 and index two; its own simplicity and identification with $\mathrm{PSU}_3(3)$ are not included in this result.

(viii) *Let F be a finite field of characteristic three with $|F| = q = 3^{2m+1}$, where $m \geq 1$. The group ${}^2G_2(F, m)$ generated by the Tits-twisted matrices in $\mathrm{GL}_7(F)$ is nonabelian simple and has order*

$$|{}^2G_2(F, m)| = q^3(q^3 + 1)(q - 1). \tag{3.3}$$

It acts faithfully and doubly transitively on a constructed set of $q^3 + 1$ points. The full point stabilizer is the soluble Borel subgroup of order $q^3(q - 1)$. Field transport

and an injective homomorphism into the split-octonion group $G_2(F)$ are proved. The parameter $q = 3$ and a graph-automorphism fixed-point characterization are not part of this assertion.

The orthogonal packages retain intrinsic kernels, full centers, root generation, singular-line geometry, and field transport. Their domains include comparison models outside the primary label ranges in Table 3. For type A and G_2, finite-field witnesses give existence at every admissible prime power; for the small Ree groups, a canonical field witness is supplied for every $m \geq 1$. The notation ${}^2G_2(F, m)$ follows the Lean model's explicit twist parameter; $|F| = 3^{2m+1}$ determines m uniquely.

Cyclic and alternating entries reuse Mathlib's principal order and simplicity results. The remaining packages combine general library theorems with the constructions described below. Mathieu parameters such as a dodecad or marked coordinates select objects in constructed finite structures; standard choices are supplied.

Formalization result 3.2 (Sporadic packages)**.** *The development contains finite nonabelian simple groups with the constructions and orders in Table 1. These entries include the specified code, lattice, stabilizer, scalar-quotient, or algebraic interfaces.*

Group	Order	Chosen construction
M_{11}	7 920	Point stabilizer in the dodecad model of M_{12}
M_{12}	95 040	Golay dodecad stabilizer
M_{22}	443 520	Pointwise stabilizer of two Golay coordinates
M_{23}	10 200 960	Stabilizer of one Golay coordinate
M_{24}	244 823 040	Full binary Golay code automorphism group
Co_1	4 157 776 806 543 360 000	Leech isometry group modulo $\{\pm I\}$
Co_2	42 305 421 312 000	Stabilizer of a Leech vector of squared norm 4
Co_3	495 766 656 000	Stabilizer of a Leech vector of squared norm 6
McL	898 128 000	Pointwise stabilizer of a Leech 2–2–3 triangle
HS	44 352 000	Pointwise stabilizer of a Leech 2–3–3 triangle
Suz	448 345 497 600	Full Eisenstein Hermitian group modulo six scalars
J_2	604 800	Full icosian Hermitian group modulo two signs
Fi_{22}	64 561 751 654 400	Common centralizer of two marked commuting involutions modulo their subgroup
Fi_{23}	4 089 470 473 293 004 800	Centralizer of a marked involution modulo that involution
Fi'_{24}	1 255 205 709 190 661 721 292 800	Parity kernel in the constructed Fischer ray group
Ru	145 926 144 000	Full Gaussian Conway–Wales lattice group modulo four scalar units

Table 1: The sixteen completed sporadic constructions. A triangle label records half the three squared edge lengths. The groups are stabilizers of vectors or configurations, with the pointwise and setwise distinctions retained.

The auxiliary groups, notably Co_0 and the retained scalar covers, are not counted as additional simple groups. Nor are small isomorphisms between family members counted as new entries. The ten sporadics outside the version 1.2 completed catalogue are $J_1, J_3, J_4, \mathrm{He}, \mathrm{O'N}, \mathrm{HN}, \mathrm{Ly}, \mathrm{Th}, B$, and the Monster. The classical extension is discussed in Section 7, and the remaining uniform Lie-type programme in Section 14. The Tits group is a separate exceptional Lie-type entry, not an additional sporadic group.

4 The enlarged development catalogue

Small derived subgroups at exceptional field sizes are not counted as additional catalogue entries when their expected isomorphism types already occur in the families. The Tits group retains its separate exceptional entry.

Relative to the eight families and fifteen sporadic constructions of version 1.0, the current development adds eight families, Rudvalis, and the derived Tits group. The assertions below concern the actual models and their completed construction checkpoints. The independent checks of 24–25 October and their remaining limits are recorded separately in Appendix C.2; they do not constitute publication of a new release.

Formalization result 4.1 (Eight additional families)**.** *For each permitted finite field size q, the retained concrete models have the following orders and are nonabelian simple in the stated ranges.*

1. *The Hermitian matrix quotient* $\mathrm{PSU}_d(q)$*, with* $d \geq 2$*, has order*

$$q^{d(d-1)/2} \prod_{i=2}^{d} (q^i - (-1)^i) / \gcd(d, q+1).$$

   Its exact nonsimple cases are $(d,q) = (2,2), (2,3), (3,2)$*. Here the catalogue label is* ${}^2A_{d-1}(q)$*.*
2. *The minus quadratic-form model* $\mathrm{P}\Omega^-_{2n}(q)$ *has order*

$$q^{n(n-1)}(q^n+1) \prod_{i=1}^{n-1} (q^{2i}-1) / \gcd(4, q^n+1).$$

   Order and simplicity are proved for all $n \geq 2$*. The stable public package uses* $n \geq 4$*, supplemented by actual rank-two* $\mathrm{PSL}_2(q^2)$ *and rank-three* $\mathrm{PSU}_4(q)$ *comparisons.*
3. *The Suzuki matrix group* ${}^2B_2(q)$*,* $q = 2^{2m+1}$*, has order* $q^2(q^2+1)(q-1)$ *for* $m \geq 0$ *and is simple exactly for* $m \geq 1$*. Its* $q = 2$ *model has order* 20 *and is not simple.*
4. *The triality matrix group* ${}^3D_4(q)$ *has order* $q^{12}(q^8+q^4+1)(q^6-1)(q^2-1)$ *and is simple for every* $q \geq 2$*.*
5. *The full split-Albert norm-and-unit stabilizer* $F_4(q)$ *has order* $q^{24}(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)$ *and is simple for every* $q \geq 2$*, including characteristics two and three.*
6. *For* $\Phi = E_6, E_7, E_8$*, the elementary adjoint Chevalley group* $G_\Phi(q)$ *is centerless and nonabelian simple over every finite field, including* $q = 2, 3$*. Its order is*

$$|G_\Phi(q)| = \frac{q^{N_\Phi} \prod_{d \in D_\Phi} (q^d - 1)}{c_\Phi(q)},$$

   with the data in Table 7. The separately constructed selected weight-companion cover has the numerator as its order and full center of order $c_\Phi(q)$*; quotienting by that center gives* $G_\Phi(q)$*.*

The unitary construction retains the full norm-one scalar center and field transport. The minus family uses the actual quadratic models in every characteristic. Suzuki and triality retain their explicit twisted root matrices, normal forms, actions and field transport; no separate full graph–field fixed-point comparison is asserted for triality or Suzuki. The F_4 proof uses the split Albert algebra, its root groups and geometric stabilizers, with separate order and simplicity audit scopes. The Chevalley construction, the E-series order data, and comparisons with earlier models are described in Section 8.

4.1 Hermitian and minus-quadratic constructions

For a finite quadratic extension K/F, conjugation $a \mapsto a^q$, where $q = |F|$, defines the standard Hermitian matrix groups $U_d(K/F)$ and $\mathrm{SU}_d(K/F)$. The projective model is the quotient of the actual SU_d by its full center: scalar matrices aI satisfying $a^{q+1} = a^d = 1$. Witt bases, norm-one stabilizers, isotropic lines, and transvections give the counting and generation arguments. The faithful primitive isotropic-line action and perfectness give simplicity in the stated range. The binary induction includes an actual $\mathrm{SU}_4(2)$ base; a proper nontrivial normal subgroup in the projective group proves the $\mathrm{PSU}_3(2)$ exception. The package retains the $\mathrm{SU}_2/\mathrm{SL}_2$ and $\mathrm{PSU}_2/\mathrm{PSL}_2$ maps, compatible field transport, and a graph–field fixed-subgroup description inside $\mathrm{SL}_d(K)$.

The minus-orthogonal form is the orthogonal sum of $n-1$ hyperbolic planes and the anisotropic norm plane of K/F. Siegel transformations generate an elementary subgroup, identified with the determinant/spinor kernel in odd characteristic and the Dickson kernel in characteristic two. Its scalar quotient is the retained $\mathrm{P}\Omega^-_{2n}$ model. The stable order and simplicity arguments for $n \geq 4$ use its own singular-line geometry and do not depend on unitary simplicity. Rank two is compared with $\mathrm{PSL}_2(K)$, over the extension field. In rank three the exterior square of K^4, with its semilinear Hodge involution, gives a six-dimensional fixed F-space carrying the minus quadratic form. The SU_4 action descends to this space; its projective kernel is exactly the full scalar center. Independent orders then make the resulting injection from PSU_4 surjective. This proves the low-rank simplicity statements without changing the stable proof.

4.2 Suzuki and triality geometry

The Suzuki model is the subgroup of $\mathrm{Sp}_4(F)$ generated by its explicit root, torus, and Weyl matrices. For $q = 2^{2m+1}$ the twist $t \mapsto t^{2^{m+1}}$ squares to Frobenius. Unique disjoint matrix cells UH and $UwHU$ give $|U| = q^2$, $|H| = q-1$, and the displayed order. The full stabilizer of infinity in the projective ovoid of size q^2+1 is the split solvable group $U \rtimes H$. Torus commutators and explicit Weyl words establish perfectness for $m \geq 1$; the faithful doubly transitive action then gives simplicity. At $m = 0$ the same order-20 model has a proper nontrivial normal Sylow 5-subgroup. Field transport commutes with the embedding in Sp_4.

For triality, take a finite cubic extension K/F. Four Frobenius-folded positive and negative simple-root families generate the actual subgroup of the split eight-dimensional quadratic isometry group over K. Positive-root coordinates $K^3 \times F^3$ give $|U| = q^{12}$; the faithful torus is $K^\times \times F^\times$. The relative Weyl quotient is dihedral of order 12, and twelve disjoint Bruhat cells give the order. The two maximal parabolics have indices $(q^3+1)(q^8+q^4+1)$ and $(q+1)(q^8+q^4+1)$. A short-root $\mathrm{SL}_2(K)$ image, the surjective cubic trace commutator, and Weyl conjugation prove perfectness in every characteristic. The faithful primitive long-parabolic action and its solvable local radical give simplicity independently of the whole-group order. The scalar intersection is trivial, giving an injection into projective elementary split $D_4(K)$. A comparison with a separately defined full graph–field fixed group remains an additional obligation.

4.3 The split Albert construction of F_4

The model is the full linear stabilizer of the cubic norm polynomial and unit on the marked 27-dimensional split Albert algebra. Preservation is an identity of polynomials, which is stronger than agreement of their values on a small finite field. There is no further scalar quotient. The 48 actual root groups generate this full stabilizer.

The trace-zero rank-one lines form a faithful primitive permutation space of size

$$\frac{(q^{12}-1)(q^4+1)}{q-1}.$$

Coordinate reconstruction gives the full point stabilizer $P = RL$ with unique factors. Its radical has order q^{15}, central additive seven-space, and additive eight-space quotient. The Levi quotient is the actual $\mathrm{SO}_{B_3}(F)$ image, with scalar kernel $F^\times$; hence

$$|L| = (q-1)q^9(q^2-1)(q^4-1)(q^6-1).$$

Orbit–stabilizer gives the order in Formalization result 4.1. Root generation, same-length A_2 commutators with coefficients ± 1, and finite stabilizer-move chains give perfectness and primitivity, including characteristics two and three. The conjugates of the local solvable radical normally generate the full group, so the solvable-radical Iwasawa criterion proves simplicity without using the final F_4 order theorem.

4.4 Rudvalis and the derived Tits group

Formalization result 4.2 (Rudvalis and Tits)**.** *The scalar quotient of the full Gaussian automorphism group of the constructed Conway–Wales lattice is nonabelian simple of order* 145 926 144 000*. The literal derived subgroup of its retained cross stabilizer is nonabelian simple of order* 17 971 200 *and embeds in the Rudvalis model with index* 8 120*. This is the retained Tits construction* ${}^2F_4(2)'$*.*

The companion source is now available as [41]. The Rudvalis construction preserves the lattice, frame geometry, scalar kernel, primitive action and exact frame sequences. The Tits package uses the inherited cross stabilizer, transfer and kernel identification, and a faithful action on 1 755 neighbours. Neither result is obtained by recognition from its order. The governing Rudvalis formalization source remains the archived Version 33 manuscript; the subsequently published paper is a bibliographic successor, not an assertion that every revision has been compared with the formalization.

The full Gaussian lattice group has order 583 704 576 000 and center exactly its four scalar units. Its quotient acts faithfully and primitively on 4 060 minimal crosses. The projective frame stabilizer is perfect of order 344 064 and, together with the glue involution, generates the quotient. Its retained exact sequences have outer quotient $\mathrm{GL}_3(2)$ and kernel of order 2^{11}; within that kernel the matrix-space quotient $M_3(\mathbb{F}_2)/\langle I\rangle$ has order 2^8 and the projected Hamming sign kernel has order 2^3. Primitivity and these local subgroups give Rudvalis simplicity without order recognition.

Write J for the actual Rudvalis cross stabilizer, of order 35 942 400. It acts faithfully and primitively on 1 755 orthogonal neighbours, with subdegrees $1, 10, 80, 640, 1024$. The computed transfer character ν has image $\{1, -1\}$, and the literal derived subgroup $T = J'$ is proved equal to $\ker \nu$. Thus $[J : T] = 2$. Local normal-subgroup arguments and the nonsquare order of T prove its simplicity. This argument uses the Rudvalis construction and order but not global Rudvalis simplicity. The restricted T action is faithful and transitive; its primitivity is not asserted. Nor is a splitting of J/T or an identification with the derived group of a separately constructed binary Ree model asserted.

Scope still open. The unfinished family labels are twisted E_6 and large Ree ${}^2F_4$; the remaining ten sporadics are listed in Section 3. The audited twisted-E_6 fixed groups and central projection are genuine construction progress, but their order, full center, and simplicity are not yet proved. Large Ree and J_1 also have partial audited results. Section 14 states the boundaries. The order-6 048 and order-504 derived-group comparisons remain unfinished.

Dependency diagrams and line counts. The latest source-bound catalogue check, on 20 October, validated 1 167 declaration references. The E-series integration supplied the three additional completed entries and their separate property audits; the later $A/B/C/D$ registration supplied additional identified realizations. Neither changes the default carriers of earlier entries. This catalogue check is distinct from independent proof replay.

The development catalogue links a self-contained HTML explorer for each completed entry, offering directed graph, shared-branch tree, exact reused declarations, and named source-theorem counts. Intrinsic lines count the union of source spans of the order and simplicity dependencies before crossing the documented reused-group boundaries. Total lines continue through all ATLAS prerequisites. Both omit Lean/Mathlib, count overlaps once per entry, and include comments and blank lines inside declaration ranges. Intrinsic does not mean mathematically exclusive: generic support and designated shared construction packages can occur in several entries. File/theorem counts count whole expanded files, while line counts count only reached declaration ranges. The development report of 18 October records the 66 principal roots, audits, source hashes and unmapped compiler declarations. Generating these reports runs no new Lean proof check. All 3,138 recorded source hashes still match this revision; no reached project declaration is unmapped.

Table 2: Development dependency footprints for all 33 completed entries, measured on 18 October 2026.

Group	Files	Theorems	Intrinsic lines	Total lines
C_p	1	18	12	12
Alt_n	1	20	7	7
$\mathrm{PSL}_d(q)$	8	82	379	379
$\mathrm{PSU}_d(q)$	82	382	4313	4658
$B_r(q)$	139	549	5653	8013
$C_r(q)$	39	177	1747	1747
$D_r(q)$	130	538	5408	5408
${}^2D_r(q)$	133	556	5966	5966
$E_6(q)$	499	5962	37707	37768
$E_7(q)$	511	6236	38713	38774
$E_8(q)$	518	6321	39417	39478
$F_4(q)$	257	1185	11416	12312
$G_2(q)$	44	290	2708	2836
${}^3D_4(q)$	83	445	3471	3494
${}^2B_2(q)$	33	139	1136	1137
${}^2G_2(q)$	46	178	2239	2250
${}^2F_4(2)'$	60	392	3080	19591
M_{11}	5	40	393	5868
M_{12}	18	89	855	5961
M_{22}	20	67	780	5351
M_{23}	5	34	397	4495
M_{24}	77	581	4183	4585
Co_1	148	766	7488	12346
Co_2	40	167	1809	14072
Co_3	85	341	3827	16423
McL	50	173	2083	16377
HS	31	146	1213	14776
Suz	212	963	11283	21374
J_2	185	803	8034	8184
Fi_{22}	36	115	1328	34250
Fi_{23}	36	115	1327	34249
Fi'_{24}	640	2509	28275	33397

Group	Files	Theorems	Intrinsic lines	Total lines
Ru	306	1777	16069	16631

5 Cyclic and alternating groups

5.1 Cyclic groups of prime order

For each prime p, the cyclic entry is the standard group C_p of order p. Its order, simplicity, and commutativity are the first assertions of Formalization result 3.1. The construction reuses Mathlib's cyclic-group results; the declaration interfaces are recorded in Appendix A. These are the commutative entries of the catalogue, kept separate from the nonabelian families.

5.2 Alternating groups

For every $n \geq 5$, the alternating entry is Alt_n, with its natural permutation action, order $n!/2$, and nonabelian simplicity. Here again the principal order and simplicity results are supplied by Mathlib. The degree convention and range are those of Formalization result 3.1; low-degree groups outside that range are not additional entries. The comparison maps with classical groups, including Alt_5, Alt_6, and Alt_8, are treated in Section 7 after the classical models have been introduced.

6 The classical series $A_r(q)$: projective special linear groups

For $A_r(q) = \mathrm{PSL}_{r+1}(q)$, write $n = r + 1$ for the matrix dimension. The projective linear construction starts from matrix groups and central quotients, followed by parameterized generation and simplicity arguments. Its Mathlib inputs are recorded in Appendix A.

6.1 Order and the determinant quotient

The starting point is the standard count of ordered bases,

$$|\mathrm{GL}_n(q)| = \prod_{i=0}^{n-1} (q^n - q^i).$$

The scalar center of $\mathrm{GL}_n(q)$ has order $q-1$. The center of $\mathrm{SL}_n(q)$ consists of the scalar matrices λI with $\lambda^n = 1$ and therefore has order $d = \gcd(n, q-1)$, by cyclicity of $\mathbb{F}_q^\times$. These counts give (3.1). The development also proves a divisibility-safe identity

$$|\mathrm{PSL}_n(q)|(q-1)d = \prod_{i=0}^{n-1} (q^n - q^i),$$

so that the natural-number division in the displayed order formula is known to be exact.

The determinant gives a stronger structural result. Over any field F and in positive dimension there is an isomorphism

$$\mathrm{PGL}_n(F)/\mathrm{PSL}_n(F) \;\cong\; F^\times/(F^\times)^n, \tag{6.1}$$

where $\mathrm{PSL}_n(F)$ on the left denotes its canonical image in $\mathrm{PGL}_n(F)$. Scaling a matrix changes its determinant by an nth power, so the map is well defined. If $\det(g) = u^n$, then $u^{-1}g$

has determinant one and represents the same projective transformation; this identifies the kernel. Surjectivity follows from that of the determinant. The formalized equivalence includes compatibility with matrix representatives. It is consequently usable independently of the finite order calculation.

6.2 Generation and simplicity

Elementary transvections are written $x_{ij}(a) = I + aE_{ij}$. The diagonal/transvection reduction available in Mathlib is completed by explicit two-coordinate identities, giving generation of $\mathrm{SL}_n(F)$ by transvections. For distinct i, j, k, the commutator identity

$$[x_{ik}(a), x_{kj}(b)] = x_{ij}(ab), \qquad [g, h] = ghg^{-1}h^{-1}, \tag{6.2}$$

proves perfectness when $n \geq 3$. Generation and perfectness pass to the projective quotient.

For a line $L = Fv$, consider the subgroup of transformations

$$u_\varphi(x) = x + \varphi(x)v, \qquad \varphi(v) = 0.$$

It is abelian, is normal in the stabilizer of L, and transforms covariantly under conjugation. Its projective images, as L varies, generate $\mathrm{PSL}_n(F)$. Together with the faithful primitive projective action, these facts fit the following criterion.

Proposition 6.1 (Iwasawa criterion)**.** *Let a nontrivial perfect group G act faithfully and primitively on a set Ω with at least two points. Suppose that for some $\alpha \in \Omega$, the stabilizer G_α has an abelian normal subgroup A whose conjugates generate G. Then G is nonabelian simple.*

Proof. Let $1 \neq N \lhd G$. The N-orbits form a G-invariant partition. Faithfulness excludes the partition into singleton orbits for such N, and primitivity therefore makes N transitive. Hence $G = NG_\alpha$. In G/N, the image of every conjugate of A is consequently the image of A itself, because $A \lhd G_\alpha$. The conjugates generate G, so G/N is abelian. Perfectness implies $G/N = 1$, and $N = G$. A nontrivial perfect group cannot be abelian. □

The existing formal criterion and projective-action infrastructure are reused. The added general-rank generation and perfectness arguments discharge its hypotheses. This proves simplicity for every field in dimension at least three. For finite fields in dimension two, the existing Mathlib theorem supplies simplicity when $q \geq 4$. The two exceptional cases are proved nonsimple using their projective actions and orders. The final family statement gives the exact exception set. Numerical examples follow by specialization.

7 The classical series B, C, D and comparison isomorphisms

The untwisted families A, B, C, D have completed packages. Following the type-A construction in Section 6, we describe the B, C, D models in that order before giving their comparisons. Throughout, $q = |F| = p^f$ for a finite field F, and r denotes *Dynkin rank*, not matrix dimension. We write Alt_m for an alternating group to distinguish it from the Lie-type symbol $A_r(q)$.

7.1 The odd-dimensional quadratic series $B_r(q)$

The quadratic construction starts from

$$Q_B(x, y, z) = \sum_{i=1}^{r} x_i y_i + z^2.$$

It constructs the subgroup generated by Siegel transformations and divides by its scalar subgroup, proved to be its full center. In the completed parameter ranges, the elementary subgroup is identified in odd characteristic with the determinant/spinor kernel. In characteristic two it is the full quadratic isometry group: the odd-dimensional space has one-dimensional polar radical, and its symplectic comparison is proved.

The geometric package has a faithful singular-line action in every positive rank, transitivity over finite fields, and primitivity from rank two. Its order is the $B_r(q)$ formula in (7.1); the exact simplicity exceptions are those in Formalization result 3.1. The odd-characteristic B_2 construction uses its symplectic C_2 image, and the geometric proof uses transport through a B_1 complement. Thus the maps $B_1 \cong A_1$ and $B_2 \cong C_2$ are part of the construction, even though the symplectic series is presented next.

7.2 The symplectic series $C_r(q)$

The model is the symplectic central quotient $C_r(F) = \mathrm{PSp}_{2r}(F)$. Its order agrees with that of $B_r(F)$, as in (7.1). For positive rank it is simple exactly outside $(r, q) = (1, 2), (1, 3), (2, 2)$ and is nonabelian at the simple parameters; the rank-zero model is trivial. The full nonsimple rank-two binary model is kept distinct from its derived subgroup.

The retained comparisons identify the rank-one model with $\mathrm{PSL}_2(F)$, the rank-two model with $B_2(F)$ over every finite field, and the higher-rank models with $B_r(F)$ in characteristic two. In odd characteristic and rank at least three, the same order does not give an isomorphism: Section 7.7 separates the two series by their involution-class counts.

7.3 The split-orthogonal series $D_r(q)$

The split quadratic construction starts from

$$Q_D(x, y) = \sum_{i=1}^{r} x_i y_i.$$

As in type B, the subgroup generated by Siegel transformations is divided by its scalar subgroup, proved to be its full center. In the completed parameter ranges it is the determinant/spinor kernel in odd characteristic and the intrinsic Dickson kernel in characteristic two. The model retains a faithful primitive singular-line action in rank at least four, together with root generation and field transport.

For every finite field and $r \geq 4$, the projective group has the order in (7.2) with $\varepsilon = 1$ and is nonabelian simple, including $D_4(2)$. The auxiliary models D_2, D_3 are retained for the product and exterior-square comparisons below; they are not additional primary family labels.

7.4 Rank conventions and order formulae

The unitary ${}^2A$ and minus-orthogonal ${}^2D$ families are completed, as described in Section 4. The common Chevalley framework and its additional realizations are described in Section 8. The superscript in ${}^2A_r$ denotes twisting; it is not a power of A_r. In twisted notation, q is the size of the fixed field: ${}^2A_r(q) = \mathrm{PSU}_{r+1}(q)$ is constructed using a Hermitian space over $\mathbb{F}_{q^2}$. References writing ${}^2A_r(q^2)$ denote the same family with a different field convention. We use the fixed-field convention also for ${}^2D_r(q)$, ${}^2E_6(q)$, and ${}^3D_4(q)$.

The symbols denote central quotients. Exceptional small parameters require conventions for Ω as in Taylor [36, Chapters 8, 10–12].

Type	Classical model	Rank range	Exclusions for simplicity
$A_r(q)$	$\mathrm{PSL}_{r+1}(q)$	$r \geq 1$	$(r,q) = (1,2), (1,3)$
$B_r(q)$	$\mathrm{P}\Omega_{2r+1}(q)$	$r \geq 2$	$(r,q) = (2,2)$
$C_r(q)$	$\mathrm{PSp}_{2r}(q)$	$r \geq 3$	None in this rank range
$D_r(q)$	$\mathrm{P}\Omega^+_{2r}(q)$	$r \geq 4$	None in this rank range
${}^2A_r(q)$	$\mathrm{PSU}_{r+1}(q)$	$r \geq 2$	$(r,q) = (2,2)$
${}^2D_r(q)$	$\mathrm{P}\Omega^-_{2r}(q)$	$r \geq 4$	None in this rank range

Table 3: Completed classical families and conventional primary rank ranges, not the full implemented parameter domains. Lower ranks are retained for comparison, and the catalogue also exposes C_2. The even-characteristic B/C comparison is proved for the actual models.

The conventional primary ranges $r \geq 1, 2, 3, 4$ for A, B, C, D remove low-rank repetitions while keeping the comparison models in the library. In particular, B, C include all positive ranks, and D_2, D_3 are available. Even-characteristic B/C isomorphisms and isolated coincidences still give overlaps between primary labels.

For reference, the classical order formulae are

$$|A_r(q)| = \frac{q^{r(r+1)/2}}{(r+1, q-1)} \prod_{i=2}^{r+1} (q^i - 1),$$

$$|B_r(q)| = |C_r(q)| = \frac{q^{r^2}}{(2, q-1)} \prod_{i=1}^{r} (q^{2i} - 1), \tag{7.1}$$

$$|D^\varepsilon_r(q)| = \frac{q^{r(r-1)}(q^r - \varepsilon)}{(4, q^r - \varepsilon)} \prod_{i=1}^{r-1} (q^{2i} - 1), \tag{7.2}$$

$$|{}^2A_r(q)| = \frac{q^{r(r+1)/2}}{(r+1, q+1)} \prod_{i=2}^{r+1} (q^i - (-1)^i).$$

Here $(a,b) = \gcd(a,b)$, $D^+_r = D_r$, and $D^-_r = {}^2D_r$ [36, 39]. All displayed formulae are proved for the actual models in the ranges stated above. The minus formula also holds at ranks two and three; the split rank-two product and rank-three exterior-square models have their separate comparisons.

7.5 Uniform coincidences and low ranks

Table 4 records the uniform identifications of central quotients, including auxiliary models below the primary rank ranges. Here ${}^2A_1$ abbreviates PSU_2; type A_1 has no nontrivial Dynkin-diagram twist.

The exterior-square representation produces the six-dimensional orthogonal model from a four-dimensional linear space. Related Klein-correspondence arguments give low-rank comparisons. In characteristic two, quotienting the polar radical of an odd-dimensional quadratic space gives a symplectic space. Over a finite field this has a converse lift [36, Chapters 11–12]. The completed untwisted comparisons implement these constructions; the minus rank-three map additionally uses semilinear Hodge descent as described in Section 4.

Isomorphism	Parameters and formalization status
$A_1(q) \cong B_1(q) \cong C_1(q) \cong \mathrm{PSU}_2(q)$	$B_1 \cong A_1$ and $C_1 \cong A_1$ over every field; $\mathrm{PSU}_2(K/F) \cong \mathrm{PSL}_2(F)$ for every finite quadratic extension. All links proved; simple for $q \geq 4$.
$B_2(q) \cong C_2(q)$	Proved over every finite field, including the nonsimple $q = 2$ case.
$D_2(q) \cong A_1(q) \times A_1(q)$	Proved over every field; the split rank-two model is not a simple entry.
$D_3(q) \cong A_3(q)$	Proved over every finite field for the split six-dimensional model.
${}^2D_2(q) \cong A_1(q^2)$	Proved for every finite quadratic extension K/F: the target is $\mathrm{PSL}_2(K)$, where $\lvert K\rvert = q^2$.
${}^2D_3(q) \cong {}^2A_3(q)$	Proved for every finite quadratic extension, through the descended exterior-square representation of $\mathrm{SU}_4(K/F)$.
$B_r(2^f) \cong C_r(2^f)$	Proved for $r \geq 1$, $f \geq 1$; primary-label overlaps persist for $r \geq 3$.

Table 4: Uniform identifications and their current proof status. For $D_2(2)$ we retain the central quotient of the two $\mathrm{SL}_2(2)$ factors, not a possibly different derived-full-orthogonal convention. The table includes comparison models outside the primary rank ranges.

7.6 Isolated coincidences across the classification list

Table 5 lists the isolated alternating and Lie-type coincidences, including simple derived groups at small parameters [5, 39, 40]. Its last column distinguishes proved maps from known isomorphisms awaiting formal comparison.

Order	Isomorphic simple groups	Status of comparison maps
60	$\mathrm{Alt}_5 \cong A_1(4) \cong A_1(5)$	Both links with Alt_5 proved.
168	$A_1(7) \cong A_2(2)$	Proved through a common eight-point action.
360	$\mathrm{Alt}_6 \cong A_1(9)$ $\cong B_2(2)' \cong C_2(2)'$	All displayed links proved; the derived-group maps are explicitly exported.
504	$A_1(8) \cong {}^2G_2(3)'$	The $q = 3$ derived-subgroup package and this comparison remain to be formalized.
6 048	${}^2A_2(3) \cong G_2(2)'$	An index-two normal subgroup of $G_2(2)$ of order 6048 is constructed; its simplicity and the unitary comparison remain open in Lean.
20 160	$\mathrm{Alt}_8 \cong A_3(2)$	Proved through the binary deleted permutation quadratic module.
25 920	${}^2A_3(2) \cong B_2(3) \cong C_2(3)$	All displayed links proved, including the unitary map through the integral E_6 Weyl representation; also ${}^2D_3(2)$.

Table 5: Isolated cross-series coincidences and their formal status, up to the uniform identifications in Table 4. A prime denotes the derived subgroup. The status column records the available comparison maps.

The binary quadratic construction gives $B_2(2) \cong \mathrm{Sym}_6$ and $B_2(2)' \cong \mathrm{Alt}_6$. The full group has order 720 and its derived group order 360. The proved B_2/C_2 map identifies the full binary symplectic model, and its restriction to the derived subgroups is exported. The separate $\mathrm{PSL}_2(9) \cong \mathrm{Alt}_6$ comparison uses the projective line and the ten partitions of six letters into two triples. Composing these proved maps completes the order-360 row.

The order-25 920 comparison uses two reductions of the same integral E_6 root lattice. Its 72 roots and reflection words give $W(E_6) \cong O_6^-(2)$, with determinant parity corresponding to Dickson parity. The determinant-one subgroup W^+ therefore identifies with the existing $\mathrm{P}\Omega_6^-(2) \cong \mathrm{PSU}_4(2)$ model. Reduction modulo three, followed by quotienting the one-dimensional radical, gives a concrete five-dimensional quadratic action. Perfectness puts its image in $B_2(3)$; a checked nonidentity image and simplicity give injectivity. The independently proved orders then give surjectivity. Composing with $B_2(3) \cong C_2(3)$ produces the actual $\mathrm{PSU}_4(2) \cong \mathrm{PSp}_4(3)$ equivalence and its transported versions. Equality of orders is used after the homomorphism and injectivity have been established. The map concerns projective groups: $\mathrm{Sp}_4(3)$ itself is a double cover of order 51 840.

For the other small parameters, $G_2(2)'$ has index two in $G_2(2)$, and ${}^2G_2(3)'$ has index three in ${}^2G_2(3)$. The Tits group ${}^2F_4(2)'$ has index two in ${}^2F_4(2)$ but gives no further coincidence in the table. The 26 sporadic groups supply no additional coincidences with each other or with the alternating and Lie-type simple groups [5, 39, 40]. Prime cyclic groups are separate; $\mathrm{Alt}_3 \cong C_3$ lies outside the chosen alternating range. These tables do not classify nonsimple groups or central covers. The comparison theorem below proves exhaustiveness only within its specified registry of completed simple models; it does not include the unfinished small derived-group packages.

7.7 Equal orders and proved nonisomorphisms

The order collisions in Section 2 require *nonisomorphism* certificates, not additional comparison isomorphisms. Write $k_2(G)$ for the number of conjugacy classes of nonidentity involutions, not the number of involutions themselves. For every finite field of odd order and every positive rank, the formalized involution-class theorems give

$$k_2(B_r(q)) = r, \qquad k_2(C_r(q)) = \lfloor r/2 \rfloor + 1. \tag{7.3}$$

Consequently, for $r \geq 3$ the constructed models satisfy

$$|B_r(q)| = |C_r(q)|, \qquad B_r(q) \ncong C_r(q).$$

The nonisomorphism follows from the isomorphism invariance of k_2.

Equal-order pair	k_2 values	Formal status
$B_r(q)$, $C_r(q)$	r, $\lfloor r/2 \rfloor + 1$	Order equality and nonisomorphism proved for odd q, $r \geq 3$.
Alt_8, $\mathrm{PSL}_3(4)$	2, 1	Nonisomorphism proved via Sylow 2-subgroup centers; these k_2 counts are not certified.

Table 6: The exceptional equal-order pairs. In the second row, the k_2 values are known mathematically, while the formal nonisomorphism uses a different invariant. Conditional on CFSG these two kinds exhaust the nonisomorphic simple-group order collisions [22, 36, 4].

The proved order-20 160 obstruction constructs the upper unitriangular subgroups in $\mathrm{SL}_4(2)$ and $\mathrm{SL}_3(4)$ and embeds them in the corresponding projective groups. Both images have order 64 and are Sylow 2-subgroups; symbolic commutation formulas show that their centers

have orders 2 and 4, respectively. An isomorphism of the ambient groups would preserve this invariant. Transport through the independently proved $\mathrm{PSL}_4(2) \cong \mathrm{Alt}_8$ map gives the alternating-group obstruction. The optional involution-class counts remain separate: Alt_8 has the two even cycle types $2^2 1^4$ and 2^4, whereas $\mathrm{PSL}_3(4)$ has one involution class [36, Chapter 4]. Both nonisomorphism routes are unconditional. Together with the proved order arithmetic and explicit equivalences, they now support the registry comparison theorem below. The conditional recognition theorem for arbitrary finite simple groups from $(|G|, k_2(G))$ remains separate.

7.8 A complete comparison theorem for the typed registry

Let $\mathcal{E}$ be the typed registry of cyclic and alternating groups, A_r, ${}^2A_r$, B_r, C_r, D_r, ${}^2D_r$, G_2, F_4, ${}^3D_4$, ${}^2B_2$, ${}^2G_2$, the sixteen sporadic constructions, and the derived Tits group. An entry retains the actual field or field extension and its chosen concrete group, not just a numerical order. Its family ranges are: prime p for C_p; $n \geq 5$ for Alt_n; $d \geq 2$ for PSL_d excluding $(2,2), (2,3)$; $d \geq 3$ for PSU_d excluding $(3,2)$; $r \geq 2$ for B_r, C_r excluding $(2,2)$; $r \geq 4$ for both signs of D_r; $q \geq 3$ for G_2; every finite field for F_4 and triality; and $m \geq 1$ for Suzuki and small Ree.

Write G_e for the actual model of $e \in \mathcal{E}$ and $\ell(e)$ for its family label and numerical parameters. Normalization identifies the proved patterns

$$\mathrm{PSL}_2(4), \mathrm{PSL}_2(5) \longmapsto \mathrm{Alt}_5, \qquad \mathrm{PSL}_3(2) \longmapsto \mathrm{PSL}_2(7), \qquad \mathrm{PSL}_2(9) \longmapsto \mathrm{Alt}_6,$$
$$\mathrm{PSL}_4(2) \longmapsto \mathrm{Alt}_8, \qquad \mathrm{PSU}_4(2) \longmapsto C_2(3), \qquad B_2(q) \longmapsto C_2(q),$$
$$B_r(q) \longmapsto C_r(q) \quad (q \text{ even},\ r \geq 3).$$

These rules are restricted to the simple parameters of the registry. All other labels are left unchanged.

Formalization result 7.1 (Comparison within the typed registry)**.** *For every $e, f \in \mathcal{E}$,*

$$G_e \cong G_f \quad \Longleftrightarrow \quad \mathrm{nf}(\ell(e)) = \mathrm{nf}(\ell(f)).$$

In particular, distinct normalized labels imply nonisomorphic groups, including the exceptional equal-order pairs.

The forward implication first uses invariance of order. The proved arithmetic classification of the registry's order formulae leaves equal normalized labels or the two collision patterns in Section 7.7. The involution-class and Sylow-center obstructions exclude the latter. Conversely, the explicit exceptional maps and compatible field-transport maps give multiplicative equivalences whenever the normalized labels agree. No CFSG hypothesis or assumed recognition certificate occurs in this theorem.

This is a result about the specified construction entries, with their infinite parameter ranges. It does not assert recognition of an arbitrary group from its order. More generally, the uniform numerical order classification combines with the proved model cardinalities to give nonisomorphism whenever the orders differ. Each model identification uses its actual multiplication-preserving equivalence.

7.9 Comparison maps and remaining targets

The comparisons marked as proved in Tables 4 and 5 are actual multiplication-preserving maps with proved inverses. Field isomorphisms induce compatible maps of the linear, Hermitian, and quadratic models, with extension data retained where required. The order-60 argument also proves an elementary recognition theorem for nonabelian simple groups of that order.

The order-6 048 and order-504 derived-group bridges remain open. For $G_2(2)$, the constructed normal subgroup of order 6 048 and index two is not yet identified with the derived subgroup or with $\mathrm{PSU}_3(3)$. The small Ree boundary at $q = 3$ likewise lacks the completed derived-group comparison with $\mathrm{PSL}_2(8)$. Comparison of central covers or actions requires the corresponding kernels and equivariance, beyond an equivalence of the projective groups.

8 Uniform Chevalley constructions and the exceptional E series

8.1 Integral root operators and central forms

The uniform construction covers $A, B, C, D, E_6, E_7, E_8, F_4, G_2$ over arbitrary finite fields. Starting from integral root data and the Geck root operators [8], it forms the positive and negative simple-root unipotents by integral divided powers before changing coefficients to the field. Thus the construction includes characteristics two and three; it does not divide by their images in the field. The adjoint model $G_\Phi(F)$ is the actual subgroup generated by these operators, with faithful adjoint linear action.

The selected weight-companion group $\widetilde{G}_\Phi(F)$ is constructed separately on the adjoint and selected weight blocks. The actual projection onto $G_\Phi(F)$ is surjective, its kernel is its full center, and the quotient equivalence is proved. These distinctions are essential: the root-generated elementary adjoint group is not asserted to be the full rational-point group of an adjoint algebraic group. Nor does the description of the cover's center claim a universal central extension or compute a Schur multiplier.

8.2 The completed E_6, E_7, E_8 families

For these three families the same carrier has the order stated in Formalization result 4.1, with the following data.

Φ	N_Φ	D_Φ	$c_\Phi(q)$
E_6	36	$2, 5, 6, 8, 9, 12$	$\gcd(3, q-1)$
E_7	63	$2, 6, 8, 10, 12, 14, 18$	$\gcd(2, q-1)$
E_8	120	$2, 8, 12, 14, 18, 20, 24, 30$	1

Table 7: Order and cover-center data for the completed E-series models.

The selected companion has order $q^{N_\Phi} \prod_{d \in D_\Phi}(q^d - 1)$, and its full center has order $c_\Phi(q)$. The adjoint catalogue carrier is centerless. It is nonabelian simple for every finite field, including all six cases $\Phi = E_6, E_7, E_8$ with $q = 2, 3$.

The order argument counts actual positive-root coordinates and disjoint Bruhat cells. Its Weyl products use the D_5 parabolic and 27-weight orbit for E_6, the E_6 parabolic and 56-weight orbit for E_7, and the E_7 parabolic and 240-root orbit for E_8. The simplicity argument uses actual normal-subgroup structure, Cartan connectivity, the core-free solvable Borel, and independently proved perfectness. Its audited dependencies exclude the whole-group order and the simplicity of an earlier realization. The paused Curtis–Tits construction is not an input. Separate property scopes tie construction, order, center, and simplicity to the same carriers and group operations.

8.3 Comparisons with the earlier models

The uniform construction supplies additional models of already completed families. Table 8 records actual comparisons; equal order and simplicity by themselves give no identification.

Type	Earlier model	Range and status
A_r	$\mathrm{PSL}_{r+1}(F)$	Every finite field, $r \geq 1$; companion identified with $\mathrm{SL}_{r+1}(F)$, and adjoint quotient with PSL.
B_r	Odd quadratic $B_r(F)$	Every finite field, $r \geq 2$, including $B_2(2)$; spin-vector map is onto with full-center kernel.
C_r	$\mathrm{PSp}_{2r}(F)$	Every finite field, $r \geq 2$, including $C_2(2)$; companion identified with $\mathrm{Sp}_{2r}(F)$.
D_r	Split quadratic $D_r(F)$	Every finite field, $r \geq 4$; natural representation and projective quotient compared explicitly.
G_2	Full split-octonion group	Every finite field, including $q = 2$; marked equivalence and ten full dependency scopes passed.
F_4	Full split-Albert group	Both models have proved order and simplicity; their equivalence remains unfinished.

Table 8: Comparison of the root-operator models with the earlier concrete groups.

The A, B, C, D maps preserve both signs of the marked simple-root generators and are registered as additional realizations. The B and C comparisons do not use either model's simplicity and therefore include the nonsimple binary rank-two case. For D, the natural representation supplies the map; new-model simplicity gives injectivity and independent orders give surjectivity, without old-model simplicity. The scalar-character kernel of the vector representation is kept distinct from the full kernel of the projective quotient.

For G_2, an integral 14-dimensional Geck module is embedded in the endomorphisms of the existing eight-dimensional octonion space, with an integral left inverse. Explicit root conjugation gives a homomorphism from a subgroup of the old model onto the new one. The independent order formulae force that subgroup to be the full old group and make the map bijective. This works at $q = 2$ without a simplicity assumption. The actual equivalence and ten full audits have passed; final preservation, indexing, and realization registration remain pending. The analogous F_4 comparison has not reached a completed theorem.

9 The exceptional family $G_2(q)$ from split octonions

The first completed exceptional Lie-type family is constructed directly from the split octonion algebra. The formal model retains the algebra, its full linear automorphism group, and its singular-point geometry; see [39, Chapter 4] for the classical background. The uniform proof includes characteristics two and three.

9.1 The algebra and its full automorphism group

Write $\mathbb{O}_{\mathrm{split}}(F) = \bigoplus_{i=1}^{8} F x_i$. The multiplication is the bilinear extension of the fixed integral split-octonion table used in the construction. In these coordinates, for $a = \sum a_i x_i$,

$$1 = x_4 + x_5, \qquad \mathrm{tr}(a) = a_4 + a_5, \qquad N(a) = a_1 a_8 + a_2 a_7 + a_3 a_6 + a_4 a_5.$$

Conjugation is $\bar{a} = \mathrm{tr}(a)1 - a$. The formal development proves both identity laws, bilinearity, the quadratic identity $a^2 - \mathrm{tr}(a)a + N(a)1 = 0$, the conjugate-product identities, norm

multiplicativity, and reversal of products under conjugation. These are polynomial identities over arbitrary commutative rings. The same algebra therefore applies in characteristic two.

The group $G_2(F)$ is the full subgroup of $\mathrm{GL}(\mathbb{O}_{\mathrm{split}}(F))$ preserving multiplication. Preservation of the identity, trace, norm, and conjugation is proved from this definition. For example, surjectivity forces the identity to be fixed; for a nonscalar a, comparison of its quadratic identity before and after applying an automorphism forces the trace and norm coefficients to agree. Thus the invariants follow from preservation of multiplication. The natural eight-dimensional representation is faithful, and coefficientwise field isomorphisms transport the group and its action.

9.2 Explicit subgroups, full stabilizers, and order

Six root transformations are defined and verified directly from the multiplication table. Their additive laws and parameter injectivity are proved. Ordered products give a bijection $F^6 \longrightarrow U$ with explicit multiplication and inverse formulas. The diagonal torus T is identified with $F^\times \times F^\times$, normalizes U, and intersects it trivially. Consequently,

$$|U| = q^6, \qquad |T| = (q-1)^2, \qquad |UT| = q^6(q-1)^2.$$

Together with explicit Weyl transformations, these maps are proved to generate the full octonion automorphism group.

The singular vectors are the nonzero a with $\mathrm{tr}(a) = N(a) = 0$. Their number is $q^6 - 1$. Indeed, after eliminating one coordinate using the trace equation, the count reduces to pairs of three-vectors and a scalar satisfying $u \cdot v = t^2$: for each $u \neq 0$ and each t, the v-fibre has q^2 elements; for $u = 0$, one has $t = 0$ and q^3 choices of v. Including zero gives q^6 vectors. The projective singular-point set $\mathcal{P}$ therefore has size

$$|\mathcal{P}| = \frac{q^6 - 1}{q - 1}.$$

Coordinate reductions prove transitivity and determine the full stabilizers. The pointwise stabilizer of the opposite vectors x_1, x_8 is isomorphic to $\mathrm{SL}_2(F)$; the stabilizer of x_1 has a parameter bijection with $F^5 \times \mathrm{SL}_2(F)$. The full stabilizer P of the line Fx_1 is the explicit parabolic, with a bijection

$$F^\times \times F^5 \times \mathrm{SL}_2(F) \;\longrightarrow\; P.$$

This bijection is a parameterization of the stabilizer. Since $|\mathrm{SL}_2(F)| = q(q^2-1)$, orbit–stabilizer gives

$$|G_2(F)| = \frac{q^6 - 1}{q - 1}\,(q-1)q^5\,q(q^2-1),$$

which is (3.2). The order proof is independent of perfectness, simplicity, and the binary exception.

9.3 Singular-point geometry and simplicity

The action on $\mathcal{P}$ is faithful. Its point stabilizer has four orbits, with lengths

$$1, \qquad q(q+1), \qquad q^3(q+1), \qquad q^5. \tag{9.1}$$

Coordinate reductions identify these strata as orbits. Their geometry also proves primitivity in every characteristic, including $q = 2$. The block-size argument leaves only the projective points of $W = \langle x_1, x_2, x_3 \rangle$ as a possible nontrivial proper block. But the linear span of products

$W \cdot W$ is the distinguished line Fx_1. Any automorphism stabilizing W must therefore fix that point, contradicting transitivity of a block stabilizer on a nontrivial block.

At Fx_1, a pair of commuting root groups gives an abelian subgroup $R_0 \cong (F^2, +)$ of order q^2, normal in the full point stabilizer. Transport defines such a subgroup at every singular point, with independence of the chosen transporter and covariance under $G_2(F)$. For $q > 2$, explicit root and Weyl relations show that their conjugates generate the full group. The argument chooses a field element distinct from 0 and 1 and applies also when $q = 3$. Torus commutators then give perfectness. The faithful primitive action and these local abelian groups satisfy Proposition 6.1, proving simplicity. Noncommutativity is established separately by two explicit root maps, even at $q = 2$. The positive simplicity proof does not use the numerical order of $G_2(F)$.

9.4 The binary exception

For $F = \mathbb{F}_2$, let H be the normal closure of the long-root groups. The construction produces a second geometry: the two-dimensional singular subspaces on which all products vanish. There are exactly 63 such zero-product planes. The count comes from an exhaustive check of 64×64 binary coordinate pairs; the group acting is still the full octonion automorphism group.

Two root involutions fix respectively 7 and 9 of these planes, so their permutation signs are $(-1)^{28} = 1$ and $(-1)^{27} = -1$. The resulting sign homomorphism is surjective and contains H in its kernel. Independently, primitivity of the singular-point action makes the nontrivial normal subgroup H transitive, and the explicit point-stabilizer generators give $[G_2(2) : H] \leq 2$. It follows that H is exactly the sign kernel. Using the already proved order,

$$|G_2(2)| = 12096, \qquad [G_2(2) : H] = 2, \qquad |H| = 6048.$$

Field transport gives the same conclusion for every field of cardinality two. This proves the negative case in the exact simplicity theorem. The present package does not prove $H = G_2(2)'$, simplicity of H, or its identification with $\mathrm{PSU}_3(3)$; the conventional derived-group coincidence in Table 5 remains a separate comparison target.

10 Small Ree groups from twisted matrices

Let F be a finite field of characteristic three with $q = |F| = 3^{2m+1}$ and $m \geq 1$. The model is the subgroup of $\mathrm{GL}_7(F)$ generated by three explicit root maps, a diagonal torus, and a Weyl involution. The coordinates follow Bäärnhielm's explicit matrix model [1, Section 3]; the geometric and simplicity arguments draw on Wilson's elementary constructions [38, 37]. Its order and simplicity are proved for this generated group, without an abstract Ree-group theorem or recognition by order.

10.1 Field twists and the root group

The twisting maps are the iterated Frobenius automorphisms

$$\theta(x) = x^{3^m}, \qquad \sigma(x) = x^{3^{m+1}}.$$

The finite-field identity $x^{3^{2m+1}} = x$ gives $\theta\sigma = \sigma\theta = \mathrm{id}$ and $\sigma^2(x) = x^3$. These identities, their compatibility with field equivalences, and the bound $q \geq 27$ are proved before the group arguments.

Write $S(a,b,c)$ for the product of the three root matrices in their specified order. Its entries recover a,b,c, and direct matrix multiplication gives

$$S(a,b,c)S(d,e,f) = S(a+d,\ b+e-a\theta(d)^3,\\ c+f-db+a\theta(d)^3d-a^2\theta(d)^3).$$

Together with the inverse formula, this identifies the root subgroup U with the parameter set F^3, so $|U|=q^3$. Projection to the first parameter has abelian image and abelian kernel, and hence U is soluble. The diagonal subgroup $H \cong F^\times$ normalizes U, with conjugation on parameters

$$(a,b,c) \longmapsto \left(\frac{a\theta(\lambda)^3}{\lambda^2}, \frac{b\lambda}{\theta(\lambda)^3}, \frac{c}{\lambda}\right).$$

Unique factorization in $B=UH$, the equality $U\cap H=1$, and the split projection $B\to H$ with kernel U prove that B is soluble and has order $q^3(q-1)$.

10.2 The point geometry and the full stabilizer

The Ree–Tits point set consists of a point at infinity and the projective classes of the first rows of the matrices $S(a,b,c)$. Projective parameter injectivity gives q^3+1 points, and U acts regularly on the affine points. The matrix convention uses row vectors; inverse right multiplication supplies the corresponding left action.

The main geometric step is to prove preservation by the Weyl element and to identify the full stabilizer of infinity. Seven exterior-kernel relations and seven Tits-twisted bilinear equations characterize the point chart. Their invariance under negative coordinate reversal proves Weyl preservation. They also imply that the last affine coordinate vanishes only at the affine origin. The generated group is consequently doubly transitive.

Every generator preserves an explicit alternating cross product and the twisted bilinear relation. At infinity the relation recovers a distinguished two-dimensional subspace, and the cross product recovers the full flag. A matrix fixing infinity must therefore be triangular. Applying the opposite flag at the affine origin shows that the two-point stabilizer is diagonal; the invariant tensors force it to be exactly H. Regularity of U then gives the full one-point stabilizer B. This proves faithfulness as well as the order formula (3.3) by orbit–stabilizer. The stabilizer calculation completes the order proof.

10.3 Perfectness, simplicity, and comparison with octonions

For $q \geq 27$, torus commutators place all three root maps in the derived subgroup. Short characteristic-three matrix identities then place the Weyl involution and the minus-one torus element there. Weyl inversion supplies all torus squares. Since $q \equiv 3 \pmod 4$, every nonzero scalar is a square or the negative of a square, so the whole torus lies in the derived subgroup. The group is therefore perfect.

Simplicity uses a separately proved general criterion: a nontrivial perfect group with a faithful quasiprimitive action and a soluble point stabilizer is simple. Indeed, any nontrivial normal subgroup N is transitive, so the stabilizer surjects onto G/N. This quotient is both soluble and perfect and is therefore trivial. The Ree action is primitive, and its full stabilizer is the soluble subgroup B. Explicit noncommuting root elements supply noncommutativity. This simplicity argument does not use the numerical group-order theorem.

Entrywise field equivalence induces a group isomorphism of the models. A separate comparison embeds the group into the split-octonion $G_2(F)$ of Section 9. In the fixed seven-

coordinate basis, the scalar–vector identification with the eight-coordinate octonions is

$$(t, v) \longmapsto (v_0, v_1, v_2, t - v_3, t + v_3, -v_4, -v_5, -v_6).$$

In characteristic three the octonion product becomes

$$(s, v)(t, w) = (st - B(v, w),\ sw + tv + v \times w),$$

where B is the preserved symmetric form and $\times$ the preserved cross product. Extending inverse row multiplication while fixing the scalar coordinate thus gives an injective homomorphism into the full octonion automorphism group. This uses structural octonion results, not the order or simplicity of $G_2(F)$. No equality with the fixed points of an exceptional graph automorphism is asserted.

Specializations of the uniform order theorem at $q = 27, 243, 2187$ give, respectively,

$$10\,073\,444\,472, \quad 49\,825\,657\,439\,340\,552, \quad 239\,189\,910\,264\,352\,349\,332\,632.$$

These values specialize the uniform theorem to constructed finite fields. The public family excludes $q = 3$; the derived-group comparison with $\mathrm{PSL}_2(8)$ in Table 5 remains a separate task.

11 Sporadic groups from retained structures

The sporadic branch follows connected constructions. The code used to construct the Mathieu groups is retained in the Leech lattice, and the resulting lattice and its full symmetry group are retained in later stabilizer and scalar-centralizer constructions. Standard background for these relationships is provided by [6]; the formal models and recorded completion results are those of the Atlas development [12]. We first specify the shared objects, then describe the individual groups in the order of Table 1. Their order and simplicity assertions are those of Formalization result 3.2.

11.1 Shared code data: trio code, hexacode, and Golay code

The initial construction passes from a quadratic alphabet and its trio code to the additive Kleinian hexacode, and then to the binary Golay code by two explicit twists. The coordinate organization

$$3 \longrightarrow 3 \times 2 \longrightarrow (3 \times 2) \times 4$$

is retained along with markings, partitions, and recovery maps. The quadratic functions on the intermediate alphabets are part of the data. Any comparison with an $\mathbb{F}_4$-linear hexacode is a separate mathematical statement.

The constructed binary code C is a doubly even self-dual $[24, 12, 8]$ code. Its weight-eight supports give the Witt design $S(5, 8, 24)$. Write $\mathcal{A} = \mathrm{Aut}(C)$ for its full coordinate automorphism group. Fix a dodecad D, a point $d \in D$, and two distinct coordinates i, j; standard choices are supplied by the construction. A subscript D denotes a setwise dodecad stabilizer, whereas coordinate subscripts denote pointwise stabilizers.

11.2 The Mathieu groups

M_{11}. The model is $(\mathcal{A}_D)_d$, the stabilizer of a point in the dodecad stabilizer. It has order 7 920. Thus the entry is specified directly by the marked Golay data; equivalently, it is the point stabilizer in the dodecad model of M_{12}.

M_{12}. The model is $\mathcal{A}_D$, the setwise stabilizer of the chosen dodecad, with its retained action on that dodecad. Its order is 95 040.

M_{22}. The model is $\mathcal{A}_{i,j}$, fixing the two marked coordinates individually, and has order 443 520. It is kept distinct from the larger setwise stabilizer of the coordinate pair.

M_{23}. The model is $\mathcal{A}_i$, the stabilizer of one marked Golay coordinate, and has order 10 200 960.

M_{24}. The model is the full coordinate automorphism group $\mathcal{A}$. Equality with the automorphism group of the octad design is proved in the same ambient symmetric group. Sextet geometry supplies local structure: every tetrad has a unique sextet completion, there are 1771 sextets, and the full sextet stabilizer has order 138240. The subsequent global action arguments give order 244823040, five-transitivity, and simplicity.

The actions on complements or dodecads, the corresponding residual designs, and the embeddings relating the marked choices are retained for the smaller Mathieu groups. Their catalogue order above does not replace the common Golay construction by an induction from smaller groups.

11.3 Shared lattice data: the Leech lattice and its full isometry group

The retained Golay code defines an integral lattice Λ. Its rank 24, evenness, positive definiteness, unimodularity, minimum squared norm 4, and generation by minimal vectors are proved from the construction. The group Co_0 is its full linear isometry group.

The shell counts for squared norms $4, 6, 8$ are respectively

$$196560, \qquad 16773120, \qquad 398034000.$$

They arise from exhaustive coordinate descriptions of lattice vectors. The global order proof uses minimal vectors and orthogonal pairs. An explicitly constructed nonmonomial isometry, together with the full monomial subgroup, generates Co_0. The resulting order is

$$|\mathrm{Co}_0| = 8\,315\,553\,613\,086\,720\,000.$$

The action on the 8292375 intrinsic crosses, and its primitivity, follow from the minimal-vector and orthogonal-pair analysis.

11.4 The Conway groups

Co_1. The simple model is the scalar quotient

$$\mathrm{Co}_1 = \mathrm{Co}_0/\langle -I \rangle.$$

It has order 4 157 776 806 543 360 000. Finiteness and the relevant kernel are proved before the final simplicity argument, so the full lattice isometry group and its simple quotient remain distinct objects.

Co_2. The model is the stabilizer in Co_0 of a Leech vector of squared norm 4. Its order is 42 305 421 312 000.

Co_3. The model is the stabilizer in Co_0 of a Leech vector of squared norm 6. Its order is 495 766 656 000; additional retained geometry includes its 276-point action.

11.5 The McLaughlin and Higman–Sims groups

McL. The model is the pointwise stabilizer of a Leech 2–2–3 triangle and has order 898 128 000. As in Table 1, the triangle label records half the three squared edge lengths.

HS. The model is the pointwise stabilizer of a Leech 2–3–3 triangle and has order 44 352 000. Its retained geometry includes the rank-three 100-point graph action.

The vector and triangle stabilizers retain their embeddings in the lattice isometry group. Their lattice representations and stabilizer descriptions allow these groups to be reused without reconstructing unrelated permutation models.

11.6 The Suzuki sporadic group Suz

The constructions of Suz and J_2 enrich the same Leech lattice with additional scalar structure. They illustrate why integral comparison maps matter: a rational isometry between two ambient spaces must be shown to map the lattices onto one another. The documented comparisons prove both integral containments.

For Suz, a constructed fixed-point-free isometry ρ satisfies

$$\rho^3 = I, \qquad \rho^2 + \rho + I = 0.$$

Its full centralizer $C_E = C_{\mathrm{Co}_0}(\rho)$ is identified with the full Hermitian isometry group of the corresponding Eisenstein lattice. Its center consists of six scalar isometries. The constructed simple quotient has order

$$|C_E|/6 = 2\,690\,072\,985\,600/6 = 448\,345\,497\,600.$$

The action on 232960 intrinsic frames and the thirteen suborbits of a frame stabilizer are proved from the lattice. The sign quotient $C_E/\{\pm I\}$ embeds into Co_1 and maps onto Suz with central kernel of order three. This keeps the embedded triple extension distinct from its simple quotient.

11.7 The Hall–Janko group J_2

For J_2, let $\mathcal{I}$ be the constructed icosian order over $\mathbb{Z}[(1+\sqrt{5})/2]$, and let C_H be the full quaternion-linear Hermitian isometry group of the associated rank-three congruence lattice. The underlying integral lattice is identified with the retained Leech lattice. The 120 norm-one icosian units act by right scalars, and C_H is their full centralizer in Co_0. The scalar action and the center of C_H are separate objects.

There are 37800 vectors of quaternionic Hermitian norm two, and each associated right quaternionic line contains exactly 120 of them. Thus there are 315 root lines. Each is orthogonal to ten others, and every orthogonal pair has a unique completing root line. This gives an intrinsic count of frames,

$$\frac{315 \cdot 10}{6} = 525.$$

The full coordinate-frame stabilizer has order 2304. Explicit quaternionic reflections generate the full group and prove frame transitivity, whence

$$|C_H| = 525 \cdot 2304 = 1209600, \qquad |C_H/\{\pm I\}| = 604800. \tag{11.1}$$

The point stabilizer has suborbits of sizes $1, 10, 32, 32, 80, 160$. Their proved geometry yields primitivity. Reflection relations yield perfectness, and the cyclic subgroup generated by a

point reflection supplies the abelian normal subgroup in Proposition 6.1. The sign quotient is therefore nonabelian simple. This example exhibits the chain from arithmetic and local geometry to a full symmetry group, its order, and its simplicity.

12 The Fischer constructions

The development contains concrete models and proofs of order, simplicity, finiteness, and noncommutativity for Fi_{22}, Fi_{23}, and Fi'_{24}. Their construction packages also retain the algebra, reflecting rays, and residue geometry described below. All three entries use one ambient ray group constructed from the algebra; the order of presentation is not an inductive construction from the smaller Fischer groups.

12.1 The common algebra and ray group

The starting point is the explicit Conway–Parker coordinate algebra constructed from Golay and Parker data [13]. The algebra has dimension 783 over $E = \mathbb{Q}(\omega)$, where ω is a primitive cube root of unity; its coordinate index is the disjoint union of the 24 Golay coordinates and the 759 octads. A proved tensor identity supplies the reflecting automorphisms. Its normalized μ_3-rays are three-element scalar orbits. Let F be the subgroup of permutations of the displayed reflecting rays generated by the constructed root involutions.

The finite-algebra construction produces three families of rays of sizes

$$24, \qquad 759 \cdot 2^5, \qquad \binom{24}{2} 2^{10},$$

whose sum is 306936. Frame geometry determines the full ray group, which has a primitive rank-three action with subdegrees $1, 31671, 275264$.

For marked commuting basic involutions t_i, a residue is obtained from the centralizer of the marked set by dividing by the subgroup it generates. The chosen marked involutions are part of the construction. The order and simplicity assertions below are proved for the resulting defined models; neither assertion is supplied as a hypothesis.

12.2 Fi_{22}

For two marked commuting basic involutions, the model is

$$G_{22} = C_F(t_i, t_j)/\langle t_i, t_j \rangle \quad (i \neq j).$$

The quotient kernel has order four. The group has a faithful primitive rank-three action on 3510 points and proved order

$$|\mathrm{Fi}_{22}| = 2^{17} 3^9 5^2 \, 7 \, 11 \, 13.$$

The construction retains the residue projection and its surjectivity, generation by distinguished involutions, and perfectness, together with the simplicity proof for this quotient.

12.3 Fi_{23}

For one marked basic involution, the model is

$$G_{23} = C_F(t_i)/\langle t_i \rangle.$$

The quotient kernel has order two. The group has a faithful primitive rank-three action on 31671 points and proved order

$$|\mathrm{Fi}_{23}| = 2^{18}3^{13}5^2\,7\,11\,13\,17\,23.$$

Here too the residue projection, its surjectivity, generation by distinguished involutions, and perfectness are retained, together with simplicity. The subdegrees of the two residue actions are

	degree	subdegrees		
G_{22}	3510	1	693	2816
G_{23}	31671	1	3510	28160

The completion record also retains the nonsplit central double cover of G_{22} inside G_{23}.

12.4 Fi'_{24}

The full three-transposition ray group F is denoted Fi_{24}. The simple model Fi'_{24} is the kernel of its parity homomorphism. This kernel is proved equal to F', with $[F : F'] = 2$, and has order

$$|\mathrm{Fi}'_{24}| = 2^{21}3^{16}5^2 7^3\,11\,13\,17\,23\,29.$$

The factor of two between $|\mathrm{Fi}_{24}|$ and $|\mathrm{Fi}'_{24}|$ is therefore part of the derived-subgroup analysis. The proved action of the simple group on the same 306936 rays is faithful and transitive; the recorded result does not include a rank-three assertion for this subgroup.

The retained perfect central triple cover of Fi'_{24} acts faithfully on the 783-dimensional E-algebra. The quotient's action is projective; no ordinary representation of the same degree is inferred for it. In the full semilinear group, odd symmetries invert the cubic scalars, so that scalar kernel is not central. Thus the full ray group, its simple derived subgroup, and the central triple cover remain distinct objects.

13 The Monster and vertex operator algebras

Höhn–Seysen [19] determines the Monster and Baby Monster orders directly from computations with axes in the 196884-dimensional Griess algebra, using Seysen's `mmgroup` implementation [33]. The Monster is generated by a subgroup of shape $2^{1+24}_{+}.\mathrm{Co}_1$ and a triality element. Orbit and stabilizer calculations use the Golay code, Leech lattice, and Conway groups.

VOA arguments complement these calculations. The axes correspond, after normalization, to Ising vectors in the Moonshine module $V^\natural$. Orbifold and fusion arguments control involutions and their centralizers, while the framed VOA structure identifies $\mathrm{Aut}(V^\natural)$ with the automorphisms of the Griess algebra preserving its invariant bilinear form. Combining the computed order with results of Carnahan and Borcherds, the paper proves that the constructed Monster is this full automorphism group and has exactly two conjugacy classes of involutions [19, Sections 2 and 5]. These are public mathematical results supporting future formalization; the Monster and Baby Monster are among the outstanding Atlas entries.

This work continues the Moonshine programme of Borcherds [2] and Frenkel–Lepowsky–Meurman [7]. Borcherds's proof of the Monstrous Moonshine conjectures [3] is a longer-term formalization goal, not a result of the present development. The Baby Monster vertex operator superalgebra provides a related source of sporadic symmetry [16]. Further examples include the 10-dimensional quadratic space over $\mathbb{F}_2$ recovered from the modules of $V^+_{\sqrt{2}E_8}$ in Shimakura's treatment, with automorphism group $O^+_{10}(2)$ [34, Theorem 4.5], and Griess–Lam's VOA existence proof of the Monster [10].

A VOA construction of the Fischer groups is being developed in joint work with Lam [18]. This manuscript in preparation is separate from the Conway–Parker coordinate-algebra source [13] used in Section 12. The latter underlies the completed Fischer formalization; the VOA construction is not included in the present release.

14 Further development

Completion of the full sporadic catalogue is a longer-term objective. The remaining ten sporadic groups in Section 3 call for geometric constructions or verified computational arguments; structural and comparison theorems for completed entries are a separate continuing task. The Hall–Janko, Fischer, Albert-algebra, and Cayley-plane source manuscripts develop the geometric direction. For O'Nan's group, Lee's 2025 account records the absence at that time of a computer-free construction [25]; Jansen–Wilson give explicit modular constructions [21]. Formalizing such arguments requires proofs of the algorithms and checks of the finite data they use.

The common Chevalley framework is now available, with completed E_6, E_7, E_8 entries and the model comparisons in Section 8. Steinberg [35] and the pinned version of Geck [8] provide mathematical sources. The remaining family targets are ${}^2E_6$ and ${}^2F_4(2^{2m+1})$; neither is counted as completed.

For twisted E_6, the diagram action has been lifted to the actual selected simply connected companion using the two dual 27-weight blocks. Over a finite quadratic extension K/F, composing it with q-Frobenius, $q = |F|$, gives the involution $x_i(t) \mapsto x_{\delta(i)}(t^q)$ on marked roots. The fixed subgroup and its surjective projection to the adjoint fixed subgroup are proved, including characteristics two and three. The restricted kernel is central of exponent dividing three, and quotienting by this actual kernel gives the adjoint fixed group. Its equality with the full center, its exact size, generation, order, and simplicity remain open. The foundations and graph lift have passed fourteen full scopes covering 184 declarations in 24 modules.

The large Ree construction retains actual root groups, torus, Weyl group, local Suzuki and linear subgroups, and a positive-root normal form of order q^{12}. Its eight affine charts give $(q^6+1)(q^3+1)(q+1)$ special projective points, including 1 755 at $q = 2$. Full stabilizer identification, full-group order, and simplicity are not yet proved. The Tits group is already constructed through Rudvalis geometry; identification with a separately defined binary Ree derived group remains unclaimed. The partial J_1 development has an actual octonion subgroup and audited perfectness, but no claimed order or simplicity theorem.

The unfinished F_4 model comparison, the small derived-group bridges, and separate graph–field fixed-point descriptions for Suzuki, triality, and small Ree are further comparison tasks. None is an assumption of the completed construction statements.

For integration with classification arguments, priorities include centralizers of prime-order elements, Sylow and local subgroups, outer automorphisms, central extensions, and recognition geometries. The comparison maps of Section 7 allow these results to pass between the models encountered in different proofs.

Formalizing the Monster and Baby Monster results discussed in Section 13 requires both verified finite computations and infrastructure for graded spaces, vertex operations, lattice extensions, and the relevant subalgebras or commutants. These developments would connect the finite-group constructions to their vertex-algebraic sources.

15 Related formalization projects

The Odd Order project of Gonthier and collaborators formalized the Feit–Thompson theorem in Coq (now Rocq), using Mathematical Components [9]. Completed in 2012 and described in 2013, it proves that finite groups of odd order are soluble and supplies extensive reusable finite-group and representation theory. Mathlib supplies the cyclic, alternating, and rank-two PSL results used here, together with group-action and simplicity infrastructure [26, 27].

The Qiuzhen-CFSG project and FormaTheoria workflow report a Lean development whose public README lists the Odd Order and Bender–Suzuki theorems as completed [31]. Their reconstruction of interdependent theories from the literature complements the construction work here. This account follows the authors' public description and does not report an independent rebuild of their project.

Independently, Yawara Ishida's `odd-order` project reports a Lean 4/Mathlib formalization of the Feit–Thompson theorem, developed from the textbooks of Isaacs, Bender–Glauberman, and Péterfalvi, together with an extensive supporting finite-group and character-theory library [20]. Its focus is the Odd Order Theorem and its prerequisites rather than a catalogue of explicit simple-group constructions. This description follows the public repository and does not report an independent rebuild.

Two Utrecht theses from 2024 precede this work: Roxy van de Kuilen's partial formalization of the first Janko group [24] and Erik van der Plas's study of the Conway groups in Lean [32]. Their precise declaration interfaces merit comparison with the constructions presented here.

The public repository `KitaKen1/finite-simple-groups-lean` reports kernel-checked order and simplicity certificates for fourteen sporadic permutation groups [23]:

$$M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, \mathrm{HS}, J_2, J_1, \mathrm{McL}, \mathrm{Co}_3, \mathrm{Suz}, \mathrm{Co}_2, \mathrm{Fi}_{22}, \mathrm{He}.$$

Its README also lists nine individual Lie-type groups, including $G_2(4)$ and $\mathrm{PSL}_3(4)$. External programs generate candidate certificates whose required properties are checked in Lean. This description follows the public repository consulted on 18 September 2026, without an independent audit of its proofs.

Parameterized families require constructions and proofs for all admissible ranks and fields. Root-group or geometric arguments provide one route; a uniformly verified algorithm could provide another. For individual groups, comparison isomorphisms could connect the permutation models with the code, lattice, and algebra models here, allowing their structural results to be combined.

A Lean implementation and mathematical interfaces

A.1 Lean and Mathlib inputs

Lean supplies dependent type theory, elaboration, and the proof kernel; Mathlib supplies mathematical definitions, theorems, and proof-producing automation. In particular, `Group`, `Nat.card`, and `IsSimpleGroup` are defined in `Mathlib.Algebra.Group.Defs`, `Mathlib.SetTheory.Cardinal.Finite`, and `Mathlib.GroupTheory.Subgroup.Simple`. Appendix B records their interpretation and the logical axioms used.

Table 9 lists representative mathematical inputs. Declaration dependencies identify the portion used by a particular proof; file imports make a broader interface available.

The cyclic entry uses `ZMod.card` and `isSimpleGroup_of_prime_card`; the alternating entry uses `nat_card_alternatingGroup` and `alternatingGroup.isSimpleGroup`. For PSL,

Representative modules (prefix `Mathlib.` omitted)	Mathematical input

`Data.Fintype.Card`
`SetTheory.Cardinal.Finite`
`GroupTheory.Index`
	Finite types, cardinalities, subgroup indices, and counting identities.

`GroupTheory.QuotientGroup.Basic`
`GroupTheory.Subgroup.Centralizer`
`GroupTheory.Commutator.Basic`
	Quotient groups, centralizers, commutators, and the maps used in central and derived-group constructions.

`GroupTheory.GroupAction.Primitive`
`GroupTheory.GroupAction.MultipleTransitivity`
`GroupTheory.GroupAction.Iwasawa`
	Primitive and multiply transitive actions and the Iwasawa simplicity criterion.

`GroupTheory.SpecificGroups.Cyclic`
`GroupTheory.SpecificGroups.Alternating`
`GroupTheory.SpecificGroups.Alternating.Simple`
	Existing cyclic and alternating group theory, including principal order and simplicity results.

`Data.ZMod.Basic` `FieldTheory.Finite.GaloisField`	Modular arithmetic and finite fields of prime-power order.

`LinearAlgebra.Matrix.GeneralLinearGroup.Card`
`LinearAlgebra.Matrix.SpecialLinearGroup`
`LinearAlgebra.Matrix.Transvection`
	Matrix groups, general-linear-group cardinalities, and elementary matrices.

`LinearAlgebra.Matrix.ProjectiveSpecialLinearGroup`
`LinearAlgebra.Projectivization.Action`
`LinearAlgebra.Projectivization.PSL.PSL2`
	Projective groups and actions, including the existing dimension-two simplicity theorem.

`LinearAlgebra.Basis.Basic`
`LinearAlgebra.BilinearForm.DualLattice`
`LinearAlgebra.QuadraticForm.Basic`
`Algebra.Quaternion`
	Bases, bilinear and quadratic forms, dual lattices, and quaternion arithmetic underlying the geometric constructions.

Table 9: Representative mathematical inputs from the pinned Mathlib dependency; they are not an exhaustive or minimal import list.

the existing `rank_two_simple` theorem and `MulAction.IwasawaStructure.isSimpleGroup` support the general-rank arguments in Section 6. Atlas develops the octonion multiplication and stabilizers, Golay–Leech structures, and Conway–Parker algebra on this foundation. Automation such as `ring` and `norm_num` produces proof terms checked by the kernel.

Shared mathematics can occur below public group interfaces. For example, McL and HS use `Atlas.Sporadic.Conway3.card`, while the Fischer entries share algebra and residue lemmas even when another Fischer group's principal wrapper is absent from their dependencies. The source-line footprints in Appendix D count Atlas declarations; their derivations also use Lean and Mathlib. The pinned environment is recorded there.

A.2 Principal declaration interfaces

The following identifiers connect the mathematical models described in Sections 5–12 to their formal statements. They are collected here to keep the construction arguments independent of implementation-level notation. The maintained `docs/catalogue/catalogue.json` and its realization profiles give the complete model/order/simplicity index for all thirty-three entries; the generated catalogue resolves the added E-series anchors.

Additional family and individual interfaces. The unitary model is `Atlas.Unitary.PSU`, with `Atlas.Unitary.card_psu_factor` and `Atlas.Unitary.simple_of_good`. The stable

minus family uses `Atlas.Orthogonal.DMinus` and `Atlas.Orthogonal.Minus.family_order`, `Atlas.Orthogonal.Minus.family_simple`; the low-rank supplement is in `Atlas/LinearGroups/Orthogonal/Minus/AllRanks.lean`. The model and principal theorems for Suzuki are `Atlas.SuzukiFamily.Group`, `card_group`, and `simple_of_large` in the same namespace; those for triality are `Atlas.Triality.Group`, `card_group`, and `simple`. For the Albert model the corresponding declarations are `Atlas.F4.Model`, `card_Model`, and `isSimple`. The Rudvalis wrappers are in `Atlas.Sporadic.Rudvalis`; the derived Tits model and principal results are `Atlas.Tits.Model`, `Atlas.Tits.card`, and `Atlas.Tits.isSimpleGroup`.

Chevalley interfaces. The carrier `Atlas.Chevalley.Model` takes the root-data tag and field. Under `Atlas.Chevalley.Catalogue`, the E_6 catalogue anchors are `E6`, `E6_order`, and `E6_simple`, with the corresponding `E7` and `E8` names. The typed finite-field domain introduces no order or simplicity assumption. Construction, full-center, quotient, and comparison declarations have separate evidence profiles. The actual octonion comparison is `Atlas.Chevalley.Comparison.G2.octonionEquiv`.

Classical comparisons. The comparisons in Tables 4 and 5 are Lean `MulEquiv` objects: multiplication-preserving maps with proved inverses. The classical index exports `b1EquivA1`, `b2EquivC2`, `d2EquivProduct`, `d3EquivA3`, and `evenBEquivC` under `Atlas.Comparisons.Classical`. The symplectic rank-one map is `Atlas.Symplectic.rankOnePSL`; field equivalences induce `bFieldEquiv` and `dFieldEquiv`. The binary comparisons are `Atlas.Orthogonal.b2BinaryEquivSymmetric` and `Atlas.Orthogonal.b2BinaryDerivedEquivAlternating`. The theorem `Atlas.Comparisons.Classical.oddB_same_order_nonisomorphic_C` records equal order and the nonexistence of a `MulEquiv`, using the conjugacy-class invariance of k_2. The order-25 920 interface is `Atlas.Comparisons.Exceptional.psu4Card2EquivPSp4Card3`. The aggregate theorem is `Atlas.Comparisons.Catalogue.isomorphic_iff_normalized_label_eq`, with corollary `Atlas.Comparisons.Catalogue.not_isomorphic_of_normalized_label_ne`. Both take actual values of `Atlas.Comparisons.Catalogue.Entry`; their scope is exactly that of Formalization result 7.1.

Split octonions. The model `Atlas.G2.Model` uses a generic construction of multiplication-preserving linear automorphisms that requires no associativity assumption. The principal declarations in `Atlas.G2` are `card_Model`, `isSimple`, and `isSimple_iff`, with `exists_mul_ne_mul` for noncommutativity. The public `Atlas.typeG2_construction` package collects the results for the concrete model, and `Atlas.typeG2_prime_power` supplies finite fields for every prime power. The corresponding existence endpoints retain the concrete model as their witness. The evidence for order, positive simplicity, the binary exception, and the complete interface is separately scoped in Appendix C. The exhaustive binary coordinate-pair check in Section 9 uses ordinary kernel reduction.

Small Ree groups. The model is `Atlas.ReeG2.ModelFm`. The principal declarations are `Atlas.ReeG2.order` and `Atlas.ReeG2.simple`. The public package `Atlas.typeReeG2_construction` and existence theorem `Atlas.exists_typeReeG2_parameter` use the same model.

Fischer groups. The declaration `Atlas.Fischer.rootGeneratedRayGroup` defines the ray group F of Section 12. The `derived` and `index` theorems identify the parity kernel with F' and prove $[F : F'] = 2$. For each Fischer entry, the public declarations `card`, `simple`, `finite`, and `exists_mul_ne_mul` establish the corresponding basic assertions. The expanded signatures contain no supplied order or simplicity hypothesis. The namespaces under `Atlas.Sporadic` are

`Fischer22`, `Fischer23`, and `Fischer24Prime`. Construction and existence interfaces accompany these results. The retained extension statements have their own statement and dependency certificates.

B Formal statements and logical foundations

For a constructed carrier G, `Group G` supplies operations and proofs of the group laws. Subgroups and quotients obtain these from their ambient groups and proved normality. On finite types, `Nat.card G` is the number of elements; it is zero on infinite types. Finiteness is also exported as `Finite G`. `IsSimpleGroup G` asserts nontriviality and that every normal subgroup is trivial or the whole group. For the nonabelian entries, noncommutativity is proved separately.

For example, the fixed model `Atlas.Sporadic.Fischer22.Model` is the centralizer quotient G_{22} of Section 12. Its public statements are

`finite` :	`Finite` G_{22},
`card` :	`Nat.card` $G_{22} = 64561751654400$,
`simple` :	`IsSimpleGroup` G_{22},
`exists_mul_ne_mul` :	$\exists x, y \in G_{22},\ xy \neq yx.$

All four declarations lie in `Atlas.Sporadic.Fischer22`; the display suppresses printer-level universe and typeclass arguments.

The recorded dependency audits report only `propext` (propositional extensionality), `Classical.choice`, and `Quot.sound` (equality of related quotient representatives). They exclude `sorryAx`, project-specific mathematical axioms, and native-evaluator trust shortcuts. The proofs have been accepted by Lean's standard kernel and, for the exported declarations specified in Appendix C.2, by the independently implemented Nanoda checker. The latest draft replay has a substantially broader scope than the historical release selection. These checks still rely on the foundations, the checking implementations, the export pipeline, and the host system. The axiom report must be read together with the full theorem type and the definitions of its carrier, parameters, and instances: an axiom check alone does not establish that the statement expresses the intended mathematics [28, 29].

C Development and recorded verification

C.1 Mathematical specification and implementation

The author specifies the objects, theorems, proof architecture, and permitted dependencies. Codex, OpenAI's coding agent, searches the pinned library and implements and repairs proofs against Lean's elaborator and kernel. The author used ChatGPT, including GPT-6 Pro, for mathematical discussion and specifications, and Codex with GPT-6 Astra for Lean implementation. These names identify the development tools used; they are not a controlled comparison of models or a measurement of autonomous performance.

The author directs and maintains Atlas and is responsible for its mathematical claims and formal specifications. Generative AI also assisted the drafting and revision of this article; the author is responsible for its final wording and mathematical content. Mathematical sources and joint work are credited where used. Compiled statements and dependency reports support review of the results; reproducing the AI's search is unnecessary for checking the resulting proof terms.

C.2 Recorded verification

The latest evidence concerns the local draft source snapshot, rather than an automatically updated public release. An incremental build, a full compiled-statement and dependency audit, an independent Nanoda replay, and a Comparator reference check address different questions. Their counts therefore need not agree.

Build and full dependency audits. The latest retained normal root build passed 8 543 Lake jobs at the twisted-E_6 graph-lift checkpoint. The catalogue check validated 1 167 declaration references. The E_6, E_7, E_8 integration has 21 separate full property scopes, with 84 root occurrences and 28 distinct declarations. The additional $A/B/C/D$ realizations have 32 full registration scopes covering 78 distinct declarations. These checks inspect complete types, instances, definitions, and transitive proof dependencies; they are not merely axiom lists.

The aggregate comparison theorem has a separate full audit of 58 roots and 79 697 dependency nodes, completed on 12 October. Its statement has the exact thirty-entry scope in Section 7.8. The construction audits retain their mathematical dependency restrictions: for example, the G_2, small Ree, Albert F_4, and E-series simplicity scopes exclude their whole-group order theorems where specified. The Tits scopes exclude global Rudvalis simplicity. These are local proof-independence claims, not claims that every construction is independent of all earlier groups.

Independent Nanoda replay: 24 October. The export covers 4 934 project modules in the normal `import Atlas` closure and 23 additional built modules. It selected 69 383 kernel declarations: 59 500 project theorems and 9 883 other declarations. Including their dependencies, Nanoda checked 139 617 **declarations with zero typechecker errors** and exited with status zero. The only permitted axioms were the three logical axioms in Appendix B.

The initial export also selected 45 compiler-generated partial implementation records; those were excluded from the kernel input, with all selected mathematical theorem roots retained and coverage checked. Fourteen source modules without successful compiled artifacts, all outside the normal root import, remain outside this certification. This is not a claim to have checked every experimental source file or all of Mathlib.

The run took about 86 minutes and reached 8.26 GiB of owned resident memory plus swap. After successful typechecking, the axiom-list pretty printer emitted a warning because it had no output destination. The original wrapper marked the run incomplete; that receipt is preserved. The supplemental `KERNEL_RESULT.json` records the successful checker exit, zero typechecker errors, unchanged source/export/binary hashes, and this reporting warning.

Comparator: 25 October. The serial run selected all 45 existing targets in 44 harnesses. **Forty-two targets passed; three remain incomplete.** Every pass includes comparison with its reference statement, the allowed-axiom check, fresh replay by Lean's standard kernel, and the quotient post-check. Comparator is not a second independent kernel implementation; Nanoda provides that separate check.

The 43 existence/discriminator targets use reviewed references whose imports are restricted to Mathlib. The two aggregate targets instead share the concrete model registry and normalization definitions with the solution; this is the explicit reference boundary for those statements. Passing an existence contract alone does not establish external recognition of a named model. Basic existence challenges no longer add a separate noncommutativity conjunct; the internal Lean theorems remain, as does cyclic commutativity. No proof rejection was reported for the three resource-stopped targets, and no full Comparator pass is claimed.

Scope	Result
Nine Chevalley contracts: $A, B, C, D, E_6, E_7, E_8, F_4, G_2$	All nine passed.
Thirty earlier existence contracts	Twenty-nine passed; the full Albert-model F_4 contract stopped at the 9 GiB resource cap.
Four strengthened discriminator contracts	All passed: the B/C involution and Alt_8 /$\mathrm{PSL}_3(4)$ Sylow-center discriminators.
Two aggregate isomorphism/nonisomorphism statements	Both incomplete: their shared harness stopped at the 9 GiB cap.

Table 10: Latest Comparator outcomes. Resource stops are neither passes nor proof rejections.

Historical version 1.0 evidence. The frozen 18 September release record remains unchanged. It reports a 5 677-job build, a full audit of 911 roots and 70 266 nodes, Nanoda acceptance of 70 891 exported declarations, and Comparator acceptance of the same 911 targets (869 theorems and 42 non-theorem equality wrappers). That older Comparator Challenge and Solution imported the same frozen release; it checked agreement with that formal reference, not an independently authored mathematical specification. The current reference contracts above are a separate run. The older source-footprint measurement in Table 11 likewise retains its date and its 46-root, 23-entry scope.

C.3 Repository and checking instructions

The public release is at `https://github.com/Moonshine-in-Kansas/atlas`; its maintained registry is `catalogue/catalogue.json`. The private development repository is `https://github.com/Moonshine-in-Kansas/atlas_draft`; it uses `docs/catalogue/catalogue.json`, with separate realization and comparison profiles. The generated `CATALOGUE.md` gives the effective completion inventory. The project guide `atlas_project.tex` describes the development workflow and requirements for new constructions.

The draft's `verification/WORKFLOWS.md` and checkpoint manifests describe full statement/dependency checks. The release's `AUDIT.md` and `verification/MAINTENANCE.md` describe its separate acceptance gate. Appendix D identifies the exact sources, tools, and receipts behind the verification claims in this note.

The release's default license is the *ATLAS Research and Attribution License 1.0*, with third-party rights preserved. The license and third-party notices accompany the source. This document revision changes neither the release nor its evidence or licensing.

D Source state and reproducibility

The current draft has principal checkout commit `95c13b57b188b350503fe3a5d6aace5f0a79047b` together with the working-tree files identified by the checker records. The commit alone does not identify all checked sources: the exact source hashes in each run are authoritative. The Nanoda and Comparator receipts report the checked mathematical sources and checker binaries unchanged at completion.

This note was revised on 25 October from the maintained catalogue, completion ledgers, comparison declarations, source-bound dependency maps, and the latest checker receipts. Its preparation involved document validation and typesetting, not another Lean build or proof replay. The development results are not automatically claims about a public release export.

The historical version 1.0 evidence concerns release commit `935474b0c3678e053e1152a49ed6593bd8ad44c9` (18 September 2026). All 2 299 recorded project-source and toolchain files matched that checked snapshot when its final Comparator and statement-inspection results were recorded. The older line measurements in Table 11 concern the development snapshot of 17 September, 07:34:33 UTC, and are not current source totals.

D.1 Source inventory

The 25 October working tree contains 4 970 Lean files under `Atlas/`, with 363 433 physical lines. Of these, 4 875 files and 355 710 lines are tracked; 95 files and 7 723 lines are untracked development sources. The root `Atlas.lean` adds 1 850 lines. These counts include comments and blank lines and do not certify every source file. In particular, the independent replay has the compiled-module scope stated in Appendix C.2.

Table 2 is the separate compiler-derived measurement of the order and simplicity dependencies of all 33 completed entries, generated on 18 October. Its 66 roots, source hashes, and group boundaries are recorded in `verification/dependency-maps/development/EVIDENCE.json`, with the row data in `COUNTS.json` in the same directory. All 3 138 source hashes still match; the later extra realizations are not silently substituted for the default models in that table. Rows share dependencies and must not be summed as disjoint contributions.

For comparison, the 17 September inventory contained 2 295 tracked Lean files under `Atlas/`, with 161 772 lines, plus the 706-line root driver: 162 478 lines in 2 296 files. This is the historical physical inventory associated with the version 1.0 discussion, distinct from either current counts or declaration-span footprints.

D.2 Build and dependency audit

The environment is pinned to Lean `leanprover/lean4:v4.34.0-rc2`, compiler commit `6a10ac8c22beadecabdbb0919c2b50214762f91d`, and Mathlib revision `85e3a25e006c35636f0e53b0e9296caca2685bc0`. From the release root, with that toolchain installed, run

```
lake exe cache get
LEAN_NUM_THREADS=2 lake build
LEAN_NUM_THREADS=2 ruby verification/check.rb
```

The first command fetches the pinned dependency cache. The remaining commands build the configured Atlas targets and run the release audit; they do not invoke Nanoda or Comparator. Preserve `lean-toolchain`, `lakefile.toml`, and `lake-manifest.json`. The release catalogue `catalogue/catalogue.json` maps claims to declarations and evidence, and `AUDIT.md` documents the build and dependency checks.

The historical version 1.0 build/audit driver ran from 05:43:04 to 05:52:59 UTC on 18 September 2026. This is the final driver's elapsed interval, not the time for a source rebuild from scratch. A preceding attempt had a process exit with status 137. The affected target was then built successfully on its own, and the final full driver passed.

D.3 Replay records and checker revisions

Current draft receipts. The Nanoda run directory is `claude/nanoda-draft-20261024/filtered-20261024T144223/`. Its `KERNEL_RESULT.json`, `SCOPE.json`, `SOURCE_HASHES.json`, `EXPORT_COVERAGE.json`, and `EXCLUDED_IMPLEMENTATIONS.json` bind the successful kernel result to its scope and source/export hashes. The original `RESULT.json` and checker log preserve the

reporting warning. The checker ran from 14:43:51 to 16:10:14 UTC on 24 October, with peak owned resident memory plus swap of 8 868 655 104 bytes.

The Comparator directory is `claude/comparator-draft-20261025/run-20261025T000133/`. Its `RESULT.json` records the complete queue, each harness's status, the resource stops, and unchanged source and binary hashes. The serial run lasted from 00:01:33 to 04:34:01 UTC on 25 October. The F_4 stop concerns the Albert-model existence target; the distinct Chevalley F_4 target passed. Both aggregate targets remain incomplete.

The E-series scopes are recorded in `verification/chevalley/integration/20261018T121057-e-series-catalogue/`; the latest catalogue evidence is `verification/runs/catalogue/20261020T020405-224/`. The full aggregate comparison audit is `verification/runs/comparison-catalogue-isomorphism/20261012T065332-1260019/`. These are dated evidence records, not instructions to relabel earlier memory stops or widen a successful run's scope.

Nanoda revision and historical replay. The checker revision is `4c544ed4099c8227f07d5de77ad1e69fb0740a27`. This same checker revision was used for the current draft. The historical release run checked 70 891 exported declarations, exited with status zero, and finished at 06:40:26 UTC on 18 September. Checking took 31 minutes 0.82 seconds on one thread, with maximum resident memory 5 375 116 KiB (about 5.13 GiB); export took 93.66 seconds. The exported data occupied 936 181 371 bytes.

Comparator revisions and historical replay. Both the historical and current runs used Comparator revision `2312244ac716564a61cc0bf4e107d9abf1757a61`, exporter revision `cacf989bd75f608700820f6afc595f32e7a99a4d`, and the Landrun sandbox at revision `811cfff51ceaf3d9843708aa6d22e9b84ccac8b4`. The historical version 1.0 run exited with status zero after 41 minutes 35.542 seconds of service runtime. Its formal-reference qualification is stated in Appendix C.2.

Statement inspections. The actual carrier applications were inspected in addition to the weaker automatic test for the presence of model declarations in type dependencies.

D.4 Per-group declaration footprints

Table 11 preserves the historical version 1.0 measurement of the selected order and simplicity declarations for its 23 entries. Its global deduplicated Atlas footprint is **88 856 lines**. The current 33-entry table is Table 2; the two selections and dates must not be conflated. This measures source dependencies, not proof-term size, model-token usage, or the amount of new mathematics.

Each column counts physical source lines in compiler-recorded declaration ranges reached through the selected theorem's transitive type and proof dependencies, including interior comments and blank lines. Generated helpers use the recorded enclosing declaration's range; a final position at column zero excludes that line. Ranges are merged by file, so column and row totals overlap. Lean core and external libraries are excluded.

For $\mathrm{PSL}_n(q)$, n is matrix dimension; r is rank for B, C, D. The type-A, symplectic, type-B, and G_2 simplicity roots are exact-range equivalences including their exceptional cases. The Ree root applies for $q = 3^{2m+1}$, $m \geq 1$. Isomorphic construction families have separate rows.

The 46-root footprint audit has 65 676 nodes, complete graph references, and resolved source ranges for every project dependency. The broader 911-declaration release audit is a separate verification scope.

Group or family	Order footprint	Simplicity footprint	Union
C_p	5	12	12
Alt_n	4	4	7
$\mathrm{PSL}_n(q)$	168	322	379
$B_r(q)$	4 121	7 898	8 009
$C_r(q)$	708	2 203	2 203
$D_r(q)$	4 290	2 466	5 408
$G_2(q)$	1 237	3 287	3 568
${}^2G_2(q)$	1 570	2 225	2 250
M_{11}	5 479	5 866	5 868
M_{12}	4 924	5 959	5 961
M_{22}	4 183	5 348	5 351
M_{23}	4 119	4 492	4 495
M_{24}	3 477	4 582	4 585
Co_1	10 826	12 334	12 346
Co_2	10 825	14 070	14 072
Co_3	12 137	16 423	16 423
McL	13 704	16 375	16 377
HS	13 447	14 771	14 776
Suz	19 727	21 320	21 374
J_2	6 989	7 954	8 184
Fi_{22}	31 955	34 248	34 250
Fi_{23}	31 950	34 247	34 249
Fi'_{24}	31 813	33 396	33 397
Global deduplicated union			**88 856**

Table 11: Historical 17 September 2026 compiler-derived Atlas source-line dependency footprints for 46 principal order and simplicity declarations: eight families and fifteen sporadic constructions. Shared source lines are counted once within each union; different rows need not have disjoint dependencies.

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